

Article

Exact Stability Atlases and a Memoryless-Surrogate Failure Theorem for a Caputo Allee Predator-Prey Model

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Abstract

We study a strong-Allee predator–prey system with Holling type-II predation under a commensurate Caputo derivative of order $\alpha \in (0, 1]$. The coexistence equilibrium is given in closed form, and its local fractional stability reduces to the trace and determinant of the Jacobian, both affine in the reciprocal Allee threshold. For every parameter set satisfying one explicit inequality, this yields a stability atlas in the (A, α) plane with three algebraic breakpoints and a critical order $\alpha^*(A)$ that decreases strictly from 1 to 0; on a benchmark slice the breakpoints are the rationals $2/23, 2/7, 10/19, 2/3$. The second result is a dichotomy about discretization. On the fractional-stabilization region, where the Caputo equilibrium is asymptotically stable despite a right-half-plane spectrum, the memoryless map $z \mapsto z + \varepsilon f(z)$ is unstable for every step size and every consistent memoryless one-step method for all small steps, whereas the implicit Grünwald–Letnikov scheme, which retains the history, is stable for every step size and reproduces the exact algebraic decay, and the explicit one for all small steps. Positivity, an extinction strip, a uniform ultimate bound, and global existence complete the picture; proof objects audit the benchmark constants and numerical corroboration covers two further parameter sets and 10^6 random ones.

Keywords: Caputo fractional derivative; strong Allee effect; Holling type-II; Matignon sector criterion; exact stability atlas; fractional stabilization; memoryless forward-Euler surrogate; Grünwald-Letnikov scheme; certified interval arithmetic; proof objects

MSC: 34A08; 34D20; 92D25; 65L20; 65G20



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1. Introduction

A strong Allee effect gives a population a threshold below which it cannot recover; a fractional-order derivative gives it memory, so that the current rate of change depends on the whole trajectory. Both ingredients are standard in population modeling. Their interaction with discretization is the subject of this paper, and it rests on a mismatch between two notions of stability. For an integer-order autonomous system, an equilibrium is linearly stable when the Jacobian spectrum lies in the open left half-plane; for a commensurate Caputo system of order α the condition is Matignon’s sector condition $|\arg \mu| > \alpha\pi/2$ [1], which for $\alpha < 1$ is strictly weaker: an eigenvalue may have a positive real part and still satisfy it. Memory can, therefore, stabilize an equilibrium that classical intuition reads as unstable. This fractional stabilization is known [2–5].

We answer two questions about it. First, where does it occur for the whole parameter family rather than for one benchmark? We show that the trace and determinant of the Jacobian at the coexistence equilibrium are affine functions of the reciprocal Allee threshold $1/A$, so that for every parameter set satisfying one explicit inequality, the (A, α) plane carries the same six-row stability atlas, with algebraic breakpoints and a critical order $\alpha^*(A)$ that decreases strictly from 1 to 0. The benchmark atlas with rational breakpoints $2/23, 2/7, 10/19, 2/3$ is one instance. Second, what does a discretization see there? If one replaces the Caputo system by the memoryless map $z \mapsto z + \varepsilon f(z)$, its multiplier at an equilibrium is $\lambda = 1 + \varepsilon \mu$ and

$$|1 + \varepsilon \mu|^2 - 1 = 2\varepsilon \operatorname{Re} \mu + \varepsilon^2 |\mu|^2,$$

so $\operatorname{Re} \mu > 0$ forces $|\lambda| > 1$ for every $\varepsilon > 0$: on the fractional-stabilization region, the map is unstable at every step size, while the continuous system is locally asymptotically stable. The identity is elementary. What is not elementary is the dichotomy it belongs to, which we prove in Section 4: every consistent memoryless one-step method inverts the verdict for all sufficiently small steps, whereas the implicit Grünwald–Letnikov scheme, which retains the history, is asymptotically stable for every step size and reproduces the exact algebraic Mittag-Leffler tail, and the explicit one is for every sufficiently small step. Refining the mesh does not repair a memoryless map, because the defect is not an accuracy defect; carrying the memory does, at any resolution.

Related Work and What Is New Here

Fractional predator–prey models with Allee effects are an active area [3–8], and comparisons of a fractional model with a discrete counterpart are established [9–12]. The numerical analysis of history-retaining fractional schemes is a developed subject: Lubich’s convolution quadratures [13], the Diethelm–Ford–Freed predictor–corrector method [14], its linear stability analysis by Garrappa [15], and the survey [16]. Validated numerics for fractional equations exist [17]. The passage from the spectrum of the Jacobian to local asymptotic stability of the nonlinear Caputo equilibrium is the rigorous linearization theorem of Cong, Doan, Siegmund, and Tuan [18], with the instability counterpart [19]; several published proofs of that step are known to be defective, so we lean on the one that has been established. We claim no novelty for fractional predator–prey dynamics with Allee effects, for order-dependent stabilization, for the existence of a critical order, for Matignon-based classification, or for the stability theory of fractional schemes. Table 1 places the closest work against the axes on which this paper differs.

The contributions are the following:

1. An exact two-scalar reduction in local fractional stability for the whole parameter family and the complete Matignon classification of the (T, D) plane (Theorems 2 and 4), together with a consolidated statement of what is and is not decided at the critical boundaries (Proposition 2).
2. A general atlas theorem: T and D are affine in $1/A$, and under the explicit open condition $c_T < 0$ every parameter set has the six-row atlas with breakpoints $A_1 < A_T < A_2 < q$ and α^* strictly decreasing (Theorem 5); the rational benchmark atlas is its instance (Theorem 6). The fractional-stabilization region is therefore an open phenomenon in parameter space, not an artifact of the benchmark.
3. A discretization dichotomy. On an analytically described region the memoryless map inverts the verdict for every $\varepsilon > 0$ (Theorem 8, Corollaries 2 and 3), and every consistent memoryless one-step method does so for all small steps (Proposition 4); the

implicit Grünwald–Letnikov scheme is verdict-faithful for every step and the explicit one for every small step (Theorem 9 and Proposition 3).

4. Global results, included for completeness and proved with standard tools: positivity and invariant strips by fractional first-contact arguments, an extinction strip that is sharp on the predator-free face, a uniform ultimate bound, and global existence obtained after the bound (Section 5), with an explicit statement of what is not proved (Proposition 5).
5. Numerical corroboration on the benchmark, on two further parameter sets, and on an ensemble of 10^6 random ones, with a convergence study near α^* and several initial conditions (Section 7); and a machine-checkable proof-object layer auditing the benchmark constants (Section 6).

Table 1. The closest related work, organized by the axes on which this paper differs. “Not reported” means the cited work does not address that axis and is a statement about scope, not a deficiency.

Reference	Model, Allee Mechanism	Derivative	Stability Criterion	Discretization Studied	Global Analysis
Wang & Han [5]	predator–prey, refuge, Allee	Caputo	Matignon	not reported	global stability
Ramesh et al. [4]	Holling I, double Allee	Caputo	Matignon	not reported	boundedness; global stability of E^*
Mondal et al. [7]	double Allee	Caputo; separate discrete-time model	Matignon; Jury	discrete-time model (flip, Neimark–Sacker)	equilibrium stability
Rahmi et al. [8]	eco-epidemiological	Caputo	Matignon	predictor–corrector simulations	global dynamics
Saha et al. [3]	Allee and fear	commensurate and incommensurate Caputo	Matignon	not reported	boundedness
Mondal, Biswas & Bairagi [12]	habitat complexity	Caputo; discretized fractional system	Matignon; Jury	order- and step-dependent discrete dynamics	global stability
Elsadany & Matouk [11]	Lotka–Volterra	Caputo; discretization	Matignon; Jury	bifurcations of the discretization	equilibrium stability
Čermák et al. [9]	linear systems	Caputo	stability regions	stability regions of discretizations	not applicable
this case	strong Allee, Holling II	Caputo	Matignon; exact affine (T, D) structure, general atlas	memoryless one-step maps against Grünwald–Letnikov: all-step dichotomy	positivity, sharp-on-face extinction strip, uniform ultimate bound

Items 2 and 3 are the new mathematics; item 1 is the reduction they rest on; item 4 is completeness; item 5 is evidence, not proof. A table at the end of Section 3 records, for every statement, whether it holds for the whole family, for the benchmark, or only as a numerical observation.

Two caveats frame everything below. The benchmark parameter values are exact rationals chosen for arithmetic convenience; they are not calibrated against any population, and we attach no ecological realism to them (Definition 1). And the paper’s practical message is about method choice, not ecology: which conclusions admit an ecological reading is discussed in Section 8.

2. Model, Nondimensionalization and Local Well-Posedness

2.1. The Model

Let ${}^C D^\alpha$ denote the Caputo derivative of order $\alpha \in (0, 1]$,

$${}^C D^\alpha g(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t (t-s)^{-\alpha} g'(s) ds, \quad 0 < \alpha < 1,$$

with ${}^C D^1 = d/dt$. We study the commensurate system

$$\begin{aligned} {}^C D^\alpha x &= P(x) - \frac{a x y}{1 + h x}, & P(x) &= r x \left(1 - \frac{x}{K}\right) \left(\frac{x}{A} - 1\right), \\ {}^C D^\alpha y &= \frac{e a x y}{1 + h x} - m y, \end{aligned} \quad (1)$$

on the closed first quadrant, with all of r, K, a, h, e, m, A positive and $0 < A < K$. Here x is prey and y predator biomass, P is a strong-Allee growth law with threshold A and carrying capacity K , predation is Holling type-II with attack rate a and handling time h , e is conversion efficiency, and m predator mortality. Writing $f = (f_1, f_2)$ for the right-hand side of Equation (1), we record two conventions and two structural facts.

The Holling term is smooth on the open half-plane

$$\Omega = \{(x, y) \in \mathbb{R}^2 : x > -1/h\}, \quad (2)$$

which already contains the closed biological quadrant, so no artificial extension of the response function is needed anywhere in this paper: all statements are made on Ω , and once positivity is proved, the solution stays a positive distance from the pole at $x = -1/h$. Two structural facts are used repeatedly:

$$f_1(0, y) = 0 \quad \text{for all } y, \quad f_2(x, 0) = 0 \quad \text{for all } x, \quad P(K) = 0. \quad (3)$$

2.2. Nondimensionalization

The Caputo derivative ${}^C D^\alpha$ carries physical dimension $T^{-\alpha}$, so comparing different values of α at fixed numerical coefficients is meaningless unless those coefficients are dimensionless. We remove the difficulty once and for all by working with a nondimensional model.

With $[x] = [K] = [A] = X$, $[y] = Y$, $[h] = X^{-1}$, $[r] = [m] = T^{-\alpha}$, $[a] = Y^{-1} T^{-\alpha}$ and $[e] = Y/X$, choose a characteristic time $t_0 > 0$ and an arbitrary predator biomass scale $Y_0 > 0$, and put

$$\tau = \frac{t}{t_0}, \quad u = \frac{x}{K}, \quad v = \frac{y}{Y_0}. \quad (4)$$

Since ${}^C D_t^\alpha = t_0^{-\alpha} {}^C D_\tau^\alpha$, substitution turns Equation (1) into

$$\begin{aligned} {}^C D_\tau^\alpha u &= R u (1 - u) \left(\frac{u}{\hat{A}} - 1\right) - \frac{\mathcal{A} u v}{1 + \eta u}, & R &= t_0^\alpha r, & \hat{A} &= A/K, & \eta &= hK, \\ {}^C D_\tau^\alpha v &= \frac{\mathcal{E} \mathcal{A} u v}{1 + \eta u} - M v, & \mathcal{A} &= t_0^\alpha a Y_0, & \mathcal{E} &= eK/Y_0, & M &= t_0^\alpha m, \end{aligned} \quad (5)$$

in which every coefficient is a pure number and, because Y_0 is free, the six effective coefficients are independent positive parameters apart from the normalisation $\hat{K} = 1$. This is Equation (1) again in the generic notation, so no generality is lost by analysing Equation (1) with dimensionless parameters throughout; in particular a benchmark with $r \neq a$ is admissible. The substitution is verified symbolically as a regression test, together with the fact that the superficially natural choice $Y_0 = r/a$ is only a *special gauge*: it forces $\mathcal{A} = R$, i.e., it selects the reduced subfamily in which the prey-growth and predation coefficients

coincide, and cannot justify a generic benchmark. An α -scan at fixed dimensionless coefficients is therefore a well-posed comparison between systems referred to the same characteristic time; it is not the same as varying α at fixed dimensional rates, about which we make no claim.

When the Coexistence Equilibrium Exists

Whenever $ea > mh$ put

$$q := \frac{m}{ea - mh}, \quad (6)$$

the prey level at which predator growth balances mortality; it does not depend on A . A positive coexistence equilibrium exists if and only if $A < q < K$ (Theorem 2), and every statement about that equilibrium below presupposes this condition.

Definition 1 (benchmark parameter set). *The benchmark is the dimensionless parameter point*

$$r = \frac{3}{2}, \quad K = 1, \quad a = 1, \quad h = \frac{1}{2}, \quad e = \frac{4}{5}, \quad m = \frac{2}{5}, \quad \text{for which } q = \frac{2}{3}. \quad (7)$$

Its values are exact rationals chosen for arithmetic convenience and reproducibility. They are not calibrated against any dataset and carry no ecological realism; every statement labeled “benchmark” is a statement about this point, and Section 3.3 shows which of its features are shared by every parameter set.

2.3. Local Well-Posedness and a Caputo Extremum Lemma

Proposition 1 (local existence, uniqueness, and regularity). *f is real-analytic on Ω ; in particular, it is locally Lipschitz there. By a solution of Equation (1) we mean, throughout, a continuous solution of the equivalent Volterra integral equation*

$$z(t) = z_0 + \frac{1}{\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} f(z(s)) ds. \quad (8)$$

For every $z_0 \in \Omega$ there is a unique such solution on a maximal interval $[0, T_{\max})$, depending continuously on z_0 (Chapter 6, [20]). Moreover, on every compact $[0, T] \subset [0, T_{\max})$ the solution satisfies

$$z \in C([0, T]) \cap C^1((0, T]), \quad \|z'(t)\| = O(t^{\alpha-1}) \text{ as } t \downarrow 0, \quad (9)$$

so that $z \in W^{1,1}(0, T)$ and, for every $t \in (0, T]$, the weighted derivative $(t-s)^{-\alpha} z'(s)$ is integrable on $(0, t)$: the Caputo derivative of each component exists pointwise on $(0, T]$ and Equation (1) holds there in the classical sense.

Proof. For $\alpha = 1$ the equation is an ordinary differential equation with real-analytic right-hand side on Ω , and the classical local theory gives a unique C^1 solution with all the stated properties; assume $0 < \alpha < 1$ from here on, so that every Beta-function statement below is in range. Existence, uniqueness, continuous dependence, and the equivalence with Equation (8) are the standard Caputo theory for a locally Lipschitz right-hand side (Chapter 6, [20]). The regularity Equation (9) is the smoothness theory for weakly singular Abel–Volterra equations, the class to which Equation (8) belongs: Lubich [21] develops the local theory and the smoothness of exact solutions, Diethelm, Ford and Freed (Theorem 2.1, [22]) record the fractional-power expansion $y(t) = \psi(t) + \sum_{\nu} c_{\nu} t^{\nu\alpha}$ with $\psi \in C^1$ for smooth data—which gives $C^1((0, T])$ and the leading $O(t^{\alpha-1})$ derivative behavior; the result is stated for a scalar equation, and the extension to the present two-dimensional system is componentwise, through the same vector-valued Volterra

regularity argument—and Diethelm [20,23] treats the Caputo setting directly. The mechanism is visible in Equation (8):

Formally,

$$z'(t) = \frac{1}{\Gamma(\alpha)} \left[t^{\alpha-1} f(z_0) + \int_0^t (t-s)^{\alpha-1} \frac{d}{ds} f(z(s)) ds \right],$$

and a weighted fixed-point argument closes this representation in the class $\{ \|z'(t)\| \leq Mt^{\alpha-1} \}$, because with $\| \frac{d}{ds} f(z(s)) \| \leq C Ms^{\alpha-1}$ the convolution obeys $\int_0^t (t-s)^{\alpha-1} s^{\alpha-1} ds = B(\alpha, \alpha) t^{2\alpha-1}$, of higher order than $t^{\alpha-1}$ near 0. Integrability of the weighted derivative follows from $\int_0^t (t-s)^{-\alpha} s^{\alpha-1} ds = B(1-\alpha, \alpha) < \infty$. This is exactly the solution class consumed by Lemma 1 below. $\square$

We do not assert global existence here: ruling out blow-up needs the a priori bound of Section 5, and asserting global existence alongside local existence would be circular. Every statement between here and Corollary 4 is therefore made on $[0, T_{\max})$ and on nothing longer.

The whole invariance argument rests on one identity, which we state and prove because the strict version of the sign is what the first-contact arguments actually consume.

Lemma 1 (Caputo extremum identity). *Let $0 < \alpha < 1$, $T > 0$, and let $w \in C([0, T]) \cap C^1((0, T])$ satisfy $w'(t) = O(t^{\alpha-1})$ as $t \downarrow 0$ — the class delivered for solution components by Proposition 1. Then for every $t \in (0, T]$ the Caputo integral converges absolutely and*

$$\Gamma(1-\alpha) {}^C D^\alpha w(t) = \frac{w(t) - w(0)}{t^\alpha} + \alpha \int_0^t \frac{w(t) - w(s)}{(t-s)^{\alpha+1}} ds. \quad (10)$$

Consequently, if $w(t_) = \max_{[0, t_*]} w$ then ${}^C D^\alpha w(t_*) \geq 0$, strictly if $w(t_*) > w(0)$; and if $w(t_*) = \min_{[0, t_*]} w$ then ${}^C D^\alpha w(t_*) \leq 0$, strictly if $w(t_*) < w(0)$.*

Proof. The stated regularity gives the weighted integrability directly: near $s = 0$, $|w'(s)| \leq Cs^{\alpha-1}$ and $\int_0^{t/2} (t-s)^{-\alpha} s^{\alpha-1} ds < \infty$, while on $[t/2, t]$ the derivative is bounded and $(t-s)^{-\alpha}$ is integrable. Hence, $(t-s)^{-\alpha} w'(s) \in L^1(0, t)$ and the Caputo integral converges absolutely. Write $w'(s) = \frac{d}{ds} (w(s) - w(t))$ and integrate by parts in $\Gamma(1-\alpha) {}^C D^\alpha w(t) = \int_0^t (t-s)^{-\alpha} w'(s) ds$:

$$\int_0^t (t-s)^{-\alpha} w'(s) ds = \left[(t-s)^{-\alpha} (w(s) - w(t)) \right]_{s=0}^{s=t} + \alpha \int_0^t \frac{w(t) - w(s)}{(t-s)^{\alpha+1}} ds.$$

The upper boundary term needs care because of the singular weight and the bounded derivative near the fixed $t > 0$ gives it at once:

$$(t-s)^{-\alpha} |w(s) - w(t)| \leq \|w'\|_{L^\infty([t/2, t])} (t-s)^{1-\alpha} \longrightarrow 0 \quad (s \uparrow t),$$

so the boundary term vanishes at $s = t$; at $s = 0$ it equals $t^{-\alpha}(w(t) - w(0))$. This is Equation (10). If $w(t_*)$ is the maximum over $[0, t_*]$ then both terms on the right of Equation (10) are nonnegative, and the first is strictly positive exactly when $w(t_*) > w(0)$. The minimum case is the same with signs reversed. The sign consequences are consistent with the classical fractional extremum principles: Luchko [24] for the original Caputo extremum principle, Luchko, Suzuki, and Yamamoto [25] for the extremum sign on the broader class $C([0, T]) \cap W^{1,1}(0, T)$, and Al-Refai [26] for related strict estimates. The pointwise identity itself is proved here, for the narrower solution class the paper actually uses, so nothing is outsourced. $\square$

Remark 1 (what is and is not available for a nonlocal derivative). A classical ODE crossing argument, which reads only the vector field at the contact point, is not available here: Cong and Tuan [27] show that whereas distinct solutions of a scalar or triangular Caputo system cannot intersect in dimension two or more two distinct solutions can meet precisely because two solutions reaching a common point may carry different histories, and Equation (1) is not triangular. What is available is a fractional first-contact argument: at a first contact the identity Equation (10) evaluates the entire history on $[0, t_*]$ and supplies the sign needed for a contradiction. That is the argument used below, and it needs no viability theorem, no change of variables and no extension of the vector field.

2.4. Positivity and Upper Bounds on the Maximal Interval

Theorem 1 (positivity and invariant upper strips). Let $0 < \alpha \leq 1$ and let the initial data be nonnegative. Then on the whole maximal interval $[0, T_{\max})$:

- (i) $x(t) \geq 0$ and $y(t) \geq 0$, and each is strictly positive whenever its initial value is;
- (ii) $x(t) \leq L := \max\{K, x(0)\}$;
- (iii) if $0 < x(0) = x_0 < A$, then $x(t) \leq x_0$.

Proof. Take first $0 < \alpha < 1$; the case $\alpha = 1$ is treated at the end.

(i) Positivity. Suppose $x(0) > 0$ and that x has a first zero at some $t_* \in (0, T_{\max})$. Then, $x(t_*) = 0 < x(0)$ and $x(t_*) = \min_{[0, t_*]} x$, so Lemma 1 gives ${}^C D^\alpha x(t_*) < 0$ strictly. But the prey equation evaluated there gives ${}^C D^\alpha x(t_*) = f_1(0, y(t_*)) = 0$ by Equation (3) — a contradiction. Hence, $x > 0$ on $[0, T_{\max})$. The same argument with $f_2(x, 0) = 0$ gives $y > 0$ when $y(0) > 0$. If $x(0) = 0$, then the pair $(0, \tilde{y})$ with ${}^C D^\alpha \tilde{y} = -m\tilde{y}$ solves Equation (1) with the same initial state, so by uniqueness $x \equiv 0$ and $y = \tilde{y} \geq 0$. If $y(0) = 0$, the face $y \equiv 0$ is likewise a solution, and the prey equation reduces to the scalar equation ${}^C D^\alpha x = P(x)$. In every case, the closed quadrant is not left.

(ii) and (iii). Both are first-contact arguments at a level $\ell + \delta$ with $P(\ell + \delta) < 0$: $\ell = L$ with $\delta > 0$ in (ii), since $L + \delta > K > A$; $\ell = x_0$ with $0 < \delta < A - x_0$ in (iii), since $0 < x_0 + \delta < A < K$. If x first attains $\ell + \delta$ at $t_\delta \in (0, T_{\max})$, then $x(t_\delta) > x(0)$ is the maximum of x over $[0, t_\delta]$ and Lemma 1 gives ${}^C D^\alpha x(t_\delta) > 0$, whereas $P(\ell + \delta) < 0$ and, by (i), the predation term is nonpositive, so the right-hand side of the prey equation is strictly negative: a contradiction. Hence $x < \ell + \delta$ on $[0, T_{\max})$ for every admissible δ , and $\delta \downarrow 0$ gives the bounds.

The endpoint $\alpha = 1$. Here, Equation (1) is an ordinary differential equation and no fractional result is invoked. Positivity is the classical quasipositivity argument — $f_1 = 0$ on $\{x = 0\}$ and $f_2 = 0$ on $\{y = 0\}$ make each axis invariant, and uniqueness keeps solutions off it — while (ii) and (iii) are the classical first-hit arguments: at a first contact with an upper level, the derivative would have to be nonnegative, whereas the right-hand side computed above is strictly negative. $\square$

3. Equilibria and the Exact Fractional Stability Atlas

3.1. Exact Equilibrium Geometry

Theorem 2 (coexistence equilibrium in closed form). Assume $ea > mh$ and let q be given by Equation (6). Then Equation (1) has a positive coexistence equilibrium if and only if $A < q < K$, in which case it is unique and equals

$$E^* = (x^*, y^*), \quad x^* = q, \quad y^* = \frac{P(q)(1 + hq)}{a q}. \quad (11)$$

Its Jacobian, simplified before any numerical substitution, is

$$J^* = \begin{pmatrix} T & -\frac{m}{e} \\ \frac{eP(q)}{q(1+hq)} & 0 \end{pmatrix}, \quad T = P'(q) - \frac{P(q)}{q(1+hq)}, \quad D = \det J^* = \frac{mP(q)}{q(1+hq)} > 0, \quad (12)$$

so the characteristic polynomial is $\lambda^2 - T\lambda + D = 0$.

Proof. On $y > 0$, $f_2 = 0$ forces $eax/(1+hx) = m$, i.e., $x(ea - mh) = m$, giving $x = q$ uniquely; $ea > mh$ makes $q > 0$. Substituting into $f_1 = 0$ and solving for y gives Equation (11). Positivity of y^* is equivalent to $P(q) > 0$, which for $0 < q < K$ holds precisely when $q > A$; and $q < K$ is needed for $1 - q/K > 0$. Hence a positive equilibrium exists iff $A < q < K$. The entries of Equation (12) follow by differentiating f and using $eaq/(1+hq) = m$ to cancel: the (2,2) entry is $eaq/(1+hq) - m = 0$, the (1,2) entry is $-aq/(1+hq) = -m/e$, and $D = -J_{12}^* J_{21}^* = mP(q)/(q(1+hq))$, positive because $P(q) > 0$. All identities in this proof, and every formula in Equations (6)–(12), were additionally verified symbolically; See Section 6. $\square$

The reduction is the point: local fractional stability of E^* now depends on the two scalars T and D only, for the whole parameter family.

Theorem 3 (boundary equilibria, with their domains of validity). *System Equation (1) has the boundary equilibria $E_0 = (0,0)$, $E_A = (A,0)$ and $E_K = (K,0)$. On $y = 0$ the Jacobian is triangular, so its eigenvalues are the diagonal entries:*

$$E_0 : \{-r, -m\}, \quad E_A : \left\{r\left(1 - \frac{A}{K}\right), \frac{eaA}{1+hA} - m\right\}, \quad E_K : \left\{-r\left(\frac{K}{A} - 1\right), \frac{eaK}{1+hK} - m\right\}.$$

Hence, E_0 is locally asymptotically stable for every $\alpha \in (0,1]$. For the benchmark Equation (7) and $0 < A < 1$:

- (i) $0 < A < 2/3$: E_A is linearly unstable, with one strictly negative and one strictly positive real eigenvalue;
- (ii) $A = 2/3$: E_A is nonhyperbolic, the second eigenvalue vanishing exactly, and the linearization decides nothing;
- (iii) $2/3 < A < 1$: E_A is linearly unstable with both eigenvalues strictly positive;

and throughout $0 < A < 2/3$, E_K is linearly unstable with one strictly negative and one strictly positive real eigenvalue, the latter equal to $2/15$.

Proof. $f_2(x,0) = 0$ gives $\partial f_2/\partial x = 0$ on $y = 0$, so the Jacobian is triangular there, and its spectrum is its diagonal. The three diagonal pairs are obtained by direct differentiation. For E_0 both entries are negative, so both eigenvalues are real negative with $\arg = \pi > \alpha\pi/2$ for every $\alpha \in (0,1]$; stability follows from [18] when $0 < \alpha < 1$ and from the classical Lyapunov indirect method at $\alpha = 1$. For the benchmark, $eaA/(1+hA) - m$ vanishes exactly at $A = 2/3$, is negative for $A < 2/3$ and positive for $A > 2/3$, while $r(1 - A/K) > 0$ for $A < 1$; this gives (i)–(iii). At $A = 4/5$, for instance, the eigenvalues of E_A are the exact rationals $3/10$ and $2/35$, both positive. For E_K , $-r(K/A - 1) < 0$ for $0 < A < K$ and $eaK/(1+hK) - m = 2/15 > 0$. $\square$

Statement (iii) corrects a claim that is easy to make and false: the eigenvalue signs at E_A are not the same for every A , and a classification of a boundary equilibrium without its A -domain is incomplete. We deliberately give eigenvalue signs rather than saddle/source language: an autonomous Caputo system does not in general generate the usual flow on

$\mathbb{R}^2$ —this is the same nonlocality recorded in Remark 1—so invariant-manifold vocabulary would assert geometry we have not proved.

3.2. The Matignon Classification

From here on E^* is the coexistence equilibrium of Theorem 2, and $T, D, \Delta = T^2 - 4D$ are its trace, determinant, and discriminant. Recall that $D > 0$ always.

Two Notions of Stability, Kept Apart

We say that E^* is Matignon-stable at order α if every eigenvalue μ of J^* satisfies $|\arg \mu| > \alpha\pi/2$, Matignon-unstable if some eigenvalue satisfies $|\arg \mu| < \alpha\pi/2$, and critical if some eigenvalue lies on the sector boundary and none strictly inside the sector. These are statements about the linearization. For $0 < \alpha < 1$, Matignon-stable implies local asymptotic stability of the nonlinear Caputo equilibrium by the linearization theorem of Cong, Doan, Siegmund, and Tuan [18], and Matignon-unstable implies instability by its counterpart [19]; at $\alpha = 1$ the classical Lyapunov indirect method gives the same two implications. Every theorem below says which notion it asserts. In tables and figures “stable” and “unstable” mean Matignon-stable and Matignon-unstable, hence also the corresponding nonlinear statements, and “critical” means that nothing about the nonlinear equilibrium is asserted.

Theorem 4 (complete Matignon classification). *Let $D > 0$ and $\alpha \in (0, 1]$. Then:*

- (a) $T < 0$: the eigenvalues lie in the open left half-plane (real or complex), so $|\arg \mu| > \pi/2 \geq \alpha\pi/2$ and E^* is locally asymptotically stable for every $\alpha \in (0, 1]$.
- (b) $T = 0$: the eigenvalues are $\pm i\sqrt{D}$, purely imaginary, and $|\arg \mu| = \pi/2$. Hence, E^* is locally asymptotically stable for every $\alpha < 1$, while at $\alpha = 1$ the linearization is critical and decides nothing.
- (c) $T > 0$ and $\Delta \geq 0$: both eigenvalues are real and positive, $\arg \mu = 0$, and E^* is unstable for every α .
- (d) $T > 0$ and $\Delta < 0$: the eigenvalues are $\rho \pm i\sigma$ with $\rho = T/2 > 0$ and $\sigma = \sqrt{4D - T^2}/2 > 0$. Setting

$$\alpha^* = \frac{2}{\pi} \arctan \frac{\sqrt{4D - T^2}}{T} \in (0, 1), \quad (13)$$

E^* is locally asymptotically stable for $0 < \alpha < \alpha^*$, unstable for $\alpha^* < \alpha \leq 1$, and at $\alpha = \alpha^*$ the linearization is critical.

Proof. Throughout, the passage from the spectrum to the nonlinear equilibrium is made in two regimes, and we keep them separate because the available theorems do. For $0 < \alpha < 1$ we use Matignon’s spectral sector criterion [1] for the linearization together with the rigorous nonlinear linearization theorem of Cong, Doan, Siegmund and Tuan [18], whose hypotheses cover exactly this range, and their instability counterpart [19] for a strict eigenvalue inside the sector. At $\alpha = 1$ neither applies—both are stated for $\alpha \in (0, 1)$ —and none is needed: the sector degenerates to the open left half-plane and the classical Lyapunov indirect method gives asymptotic stability when every eigenvalue has a strictly negative real part, and instability when one has a strictly positive real part.

$D > 0$ excludes a real pair of opposite signs, so real eigenvalues share the sign of T . Cases (a) and (c) are then immediate from $\arg \mu \in \{0, \pi\}$ for real eigenvalues, in both regimes. In (b), $T = 0$ with $D > 0$ gives $\lambda^2 = -D$, so $\mu = \pm i\sqrt{D}$ and $|\arg \mu| = \pi/2$; the sector condition $\pi/2 > \alpha\pi/2$ holds iff $\alpha < 1$, and at $\alpha = 1$ we have equality, where the linear theorem is silent. In (d), $\arg \mu = \arctan(\sigma/\rho) \in (0, \pi/2)$ because $\rho, \sigma > 0$, and the sector condition

$\arctan(\sigma/\rho) > \alpha\pi/2$ is, since $\tan$ is increasing on $(0, \pi/2)$, equivalent to $\alpha < (2/\pi) \arctan(\sigma/\rho)$, which is Equation (13) after substituting ρ and σ . $\square$

The boundary cases deserve one consolidated statement, because a change of linear classification is easily mistaken for a bifurcation theorem.

Proposition 2 (what is and is not decided at the critical boundaries). *Let $D > 0$.*

- (i) *At $T = 0$ with $\alpha = 1$, and at $\alpha = \alpha^*$ in case (d), the eigenvalue pair lies exactly on the sector boundary $|\arg \mu| = \alpha\pi/2$. The linearization is critical: neither [18] nor [19] applies, and nothing about the nonlinear equilibrium is asserted there in this paper.*
- (ii) *Crossing $\alpha = \alpha^*(A)$, or crossing $T = 0$ at $\alpha = 1$, is a crossing of a linear stability boundary. It is not asserted to be a Hopf bifurcation, and no periodic orbit is claimed. For $0 < \alpha < 1$ exact nonconstant periodic solutions are excluded for broad classes of autonomous Caputo systems [28,29]; long transients and asymptotically periodic behavior are separate questions, not addressed here.*
- (iii) *At $\Delta = 0$ the eigenvalue is a double real one of the sign of T ; the verdict is decided and α -independent: Matignon-stable if $T < 0$, Matignon-unstable if $T > 0$.*
- (iv) *At $A = q$ the coexistence equilibrium reaches $y^* = 0$ and ceases to exist, while E_A acquires a zero eigenvalue. This is an existence and nonhyperbolicity boundary, not an unresolved verdict about an interior equilibrium.*

Proof. (i) is the equality case of Theorem 4(b) and (d); both cited theorems assume a strict inequality. (ii) is the definition of the boundary together with the cited results. (iii): $\Delta = 0$ with $D > 0$ gives a double real root of $\lambda^2 - T\lambda + D$ with the sign of T , so $\arg \mu \in \{0, \pi\}$ and Theorem 4 (a) or (c) applies. (iv): $y^*(A) = P(q)(1 + hq)/(aq) \rightarrow 0$ as $A \uparrow q$, and the second eigenvalue of E_A in Theorem 3, $eaA/(1 + hA) - m$, vanishes exactly when $A(ea - mh) = m$, i.e., at $A = q$. $\square$

3.3. The Atlas for an Arbitrary Parameter Set

The benchmark is not special. Fix a parameter set with $ea > mh$ and $q < K$, write $u = 1/A$, and put

$$s := 1 - \frac{q}{K} \in (0, 1), \quad w := 1 + hq. \quad (14)$$

Theorem 5 (structure of the atlas for every parameter set). *Let $ea > mh$ and $q < K$, so that the coexistence equilibrium exists exactly for $A \in (0, q)$.*

- (a) *T and D are affine in $u = 1/A$:*

$$\begin{aligned} T &= c_T u + d_T, & D &= c_D u + d_D, \\ c_T &= rq \left(2 - \frac{3q}{K} - \frac{s}{w} \right), & d_T &= r \left(\frac{2q}{K} + \frac{s}{w} - 1 \right), \\ c_D &= \frac{mrqs}{w}, & d_D &= -\frac{mrs}{w}. \end{aligned} \quad (15)$$

Hence $D = (mrs/w)(q/A - 1) > 0$ on $(0, q)$, $T(q) = rs > 0$, and Δ is a quadratic in u with leading coefficient c_T^2 . The classification of Theorem 4 changes at no more than three values of A in $(0, q)$, and on a left neighborhood of $A = q$ the eigenvalues are real and positive.

- (b) *If $c_T < 0$, equivalently*

$$2 - \frac{3q}{K} < \frac{1 - q/K}{1 + hq}, \quad (16)$$

then T has exactly one zero $A_T = -c_T/d_T \in (0, q)$, Δ has exactly two zeros $0 < A_1 < A_T < A_2 < q$, and the six-row classification of Table 2 holds verbatim

with $(2/23, 2/7, 10/19, 2/3)$ replaced by (A_1, A_T, A_2, q) : a real negative pair on $(0, A_1]$, a complex left-half-plane pair on (A_1, A_T) , a purely imaginary pair at A_T , a complex right-half-plane pair on (A_T, A_2) and a real positive pair on $[A_2, q)$.

- (c) Under (b), α^* is strictly decreasing on (A_T, A_2) , with $\alpha^*(A) \rightarrow 1$ as $A \downarrow A_T$ and $\alpha^*(A) \rightarrow 0$ as $A \uparrow A_2$. The fractional-stabilization region

$$\mathcal{W} = \{(A, \alpha) : A_T < A < A_2, 0 < \alpha < \alpha^*(A)\} \quad (17)$$

is nonempty with nonempty interior, and on it E^* is locally asymptotically stable although $\operatorname{Re} \mu = T/2 > 0$.

- (d) If $c_T \geq 0$ then $T > 0$ on all of $(0, q)$. Fractional stabilization occurs exactly on $\{A \in (0, q) : \Delta(A) < 0\}$, an interval that may be empty, and there $\alpha^* < 1$.

Table 2. The benchmark stability atlas of Theorem 6. “Stable” and “unstable” are Matignon-stable and Matignon-unstable, hence also locally asymptotic stability and instability of the nonlinear equilibrium for $0 < \alpha < 1$; “critical” asserts nothing. By Theorem 5 (b) the same six rows hold for every parameter set with $c_T < 0$, with breakpoints $A_1 < A_T < A_2 < q$.

Range of A	General Symbol	Eigenvalue Geometry	Caputo Verdict
$0 < A \leq 2/23$	$(0, A_1]$	real negative (double at A_1)	stable for all $\alpha \in (0, 1]$
$2/23 < A < 2/7$	(A_1, A_T)	complex, left half-plane	stable for all $\alpha \in (0, 1]$
$A = 2/7$	A_T	purely imaginary	stable for $\alpha < 1$; critical at $\alpha = 1$
$2/7 < A < 10/19$	(A_T, A_2)	complex, right half-plane	stable iff $\alpha < \alpha^*(A)$
$10/19 \leq A < 2/3$	$[A_2, q)$	real positive (double at A_2)	unstable for all α
$A \geq 2/3$	$[q, \infty)$	no coexistence equilibrium	not applicable

Proof. (a) By Equations (11)–(12), $P(q) = rqs(qu - 1)$ and $P'(q) = r[(2q - 3q^2/K)u - 1 + 2q/K]$, so $T = P'(q) - P(q)/(qw)$ and $D = mP(q)/(qw)$ are the affine functions displayed, with the stated coefficients. At $u = 1/q$ the value of T is $r(2 - 3q/K - s/w) + r(-1 + 2q/K + s/w) = r(1 - q/K) = rs$, and $D(q) = 0$. Since $\Delta = (c_T u + d_T)^2 - 4(c_D u + d_D)$, it is a quadratic in u with leading coefficient c_T^2 ; its zeros and the zero of T are the only points where the case of Theorem 4 can change. As $A \uparrow q$, $D \rightarrow 0^+$ and $T \rightarrow rs > 0$, so $\Delta \rightarrow r^2 s^2 > 0$: a real positive pair.

(b) If $c_T < 0$ then T is strictly decreasing in u with $T(1/q) = rs > 0$ and $T \rightarrow -\infty$ as $u \rightarrow \infty$, so it has exactly one zero $u_T > 1/q$, i.e., $A_T = 1/u_T = -c_T/d_T \in (0, q)$; here $d_T = rs - c_T/q > 0$. At u_T , $\Delta = -4D < 0$. The quadratic $\Delta(u)$ is convex, negative at u_T , positive at $u = 1/q$, and tends to $+\infty$ as $u \rightarrow \infty$; hence it has exactly one zero $u_2 \in (1/q, u_T)$ and one zero $u_1 > u_T$. With $A_i = 1/u_i$ this gives $0 < A_1 < A_T < A_2 < q$, $\Delta < 0$ exactly on (A_1, A_2) and $T < 0$ exactly on $(0, A_T)$; the rows follow from Theorem 4 with $D > 0$.

(c) Write $\alpha^* = (2/\pi) \arctan \sqrt{R}$ with $R = (4D - T^2)/T^2$, as in Equation (13). Differentiating in u ,

$$\frac{dR}{du} = \frac{4(c_D T - 2D c_T)}{T^3},$$

and on (A_T, A_2) we have $T > 0$, $D > 0$, $c_D > 0$ and $c_T < 0$, so $dR/du > 0$: R increases with u , decreases with A , and so does α^* . As $A \downarrow A_T$, $T \rightarrow 0^+$ with $D > 0$, so $R \rightarrow \infty$ and $\alpha^* \rightarrow 1$; as $A \uparrow A_2$, $\Delta \rightarrow 0^-$, so $R \rightarrow 0^+$ and $\alpha^* \rightarrow 0$. Local asymptotic stability on $\mathcal{W}$ is Theorem 4 (d).

(d) If $c_T \geq 0$ then T is nondecreasing in u and $T(1/q) > 0$, so $T > 0$ on $(0, q)$. A complex pair occurs exactly where the convex quadratic Δ is negative, which is an interval, and there $\arg \mu \in (0, \pi/2)$ gives $\alpha^* < 1$. Every identity in this proof was additionally verified symbolically. $\square$

Corollary 1 (the fractional-stabilization region is an open phenomenon). *The set of parameter vectors $(r, K, a, h, e, m) \in (0, \infty)^6$ with $ea > mh$, $q < K$ and $c_T < 0$ is open, and the benchmark lies in it with $(c_T, d_T, c_D, d_D) = (-\frac{1}{4}, \frac{7}{8}, \frac{1}{10}, -\frac{3}{20})$. Hence, every parameter set in a neighborhood of the benchmark has the six-row atlas and a nonempty region Equation (17). Condition Equation (16) holds whenever $q/K \geq 2/3$, and as $h \rightarrow 0$ it reduces to $q/K > 1/2$: the coexistence prey level must exceed roughly half the carrying capacity.*

Proof. c_T is a continuous function of the parameters on the open set $\{ea > mh, q < K\}$, so $\{c_T < 0\}$ is open; the benchmark values are a direct evaluation of Equation (15). If $q/K \geq 2/3$ then the left side of Equation (16) is nonpositive and the right side is positive. As $h \rightarrow 0$ the right side tends to $1 - q/K$, and $2 - 3q/K < 1 - q/K$ iff $q/K > 1/2$. $\square$

Section 7.5 exhibits the theorem on two further parameter sets, one of them with irrational coefficients, and quantifies on an ensemble of random parameter sets how often Equation (16) holds.

3.4. The Benchmark Slice

Theorem 6 (exact atlas on the benchmark slice). *For the benchmark Equation (7), $q = x^* = 2/3$ independently of A , and*

$$\begin{aligned} y^*(A) &= \frac{2(2-3A)}{9A}, & T(A) &= \frac{7A-2}{8A}, \\ D(A) &= \frac{2-3A}{20A}, & \Delta(A) &= \frac{(19A-10)(23A-2)}{320A^2}. \end{aligned} \quad (18)$$

A positive coexistence equilibrium exists exactly for $0 < A < 2/3$, and on that range the classification is the one of Table 2, with $\alpha^(A)$ given by Equation (13). All four breakpoints are exact rationals, and $(c_T, d_T, c_D, d_D) = (-\frac{1}{4}, \frac{7}{8}, \frac{1}{10}, -\frac{3}{20})$ in Equation (15).*

Proof. Substituting Equation (7) in Equation (6) gives $q = (2/5)/(4/5 - 1/5) = 2/3$, independent of A ; Equation (18) then follows from Equations (11) and (12) by direct rational computation. The zeros of the numerators locate the breakpoints: $y^* > 0$ iff $A < 2/3$; $T = 0$ iff $A = 2/7$; and $\Delta = 0$ iff $A \in \{2/23, 10/19\}$, with $\Delta < 0$ strictly between them. Since $2/23 < 2/7 < 10/19 < 2/3$, the six rows are exactly the cases of Theorem 4 on the successive subintervals. Each identity in Equation (18) is an equality of rational functions and was verified symbolically; the values at the four breakpoints are additionally carried by machine-checkable objects (Section 6). $\square$

Theorem 7 (strict monotonicity of the critical order). *On $(2/7, 10/19)$ define $R(A) = (4D - T^2)/T^2$. For the benchmark slice*

$$R(A) = -\frac{(19A-10)(23A-2)}{5(7A-2)^2}, \quad R'(A) = -\frac{128(A+2)}{5(7A-2)^3}. \quad (19)$$

On $(2/7, 10/19)$ we have $7A-2 > 0$ and $A+2 > 0$, so $R' < 0$. Since $\alpha^ = (2/\pi) \arctan \sqrt{R}$ and both $\sqrt{\cdot}$ and $\arctan$ are strictly increasing, α^* is strictly decreasing on $(2/7, 10/19)$, with*

$$\lim_{A \downarrow 2/7} \alpha^*(A) = 1, \quad \lim_{A \uparrow 10/19} \alpha^*(A) = 0.$$

Proof. This is Theorem 5 (c) at $c_T = -1/4 < 0$. The explicit forms Equation (19) are equalities of rational functions obtained by substituting Equation (18); they are recorded because the proof objects of Section 6 audit them, and they were verified symbolically. $\square$

Figure 1 is drawn from the closed form; no sampled grid establishes any boundary in it.

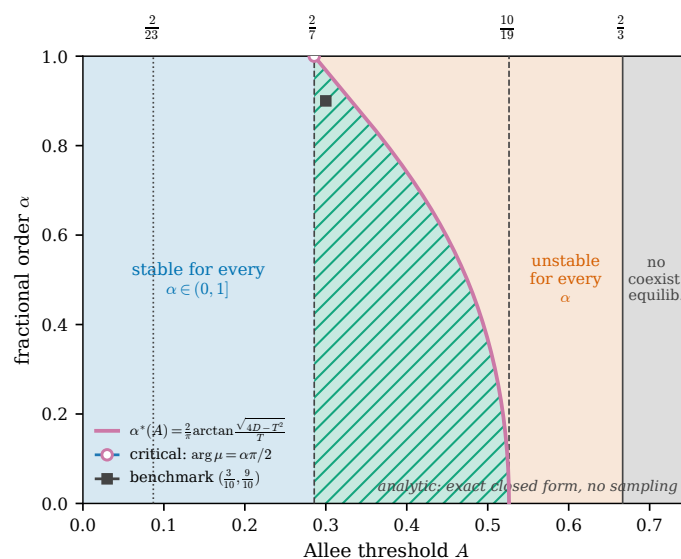


Figure 1. The benchmark atlas of Theorem 6 from the closed form Equation (13); landmarks at $2/23$, $2/7$, $10/19$, $2/3$. Hatched: the region $\mathcal{W}$ of Corollary 2, where E^* is stable and the memoryless map is unstable for every $\varepsilon > 0$. Open circle: the critical point of Proposition 2 (i). Analytic content.

Remark 2 (the benchmark point). At $A = 3/10$ one gets the exact values $T = 1/24$, $D = 11/60$, $\Delta = -2107/2880 < 0$, so case (d) applies with

$$\mu_{\pm} = \frac{1}{48} \pm i \frac{1}{2} \sqrt{\frac{2107}{2880}}, \quad \alpha^*(3/10) = 0.969012276152 \dots$$

In particular $\operatorname{Re} \mu = 1/48 > 0$: classical linear intuition reads this equilibrium as unstable. Since the benchmark order is $\alpha = 9/10 < \alpha^*(3/10)$, the Caputo equilibrium is in fact locally asymptotically stable. This single point is an illustration; the content is the region.

Table 3 records the scope of every statement in the paper.

Table 3. Scope of every statement in the paper: what it asserts, for which parameters, and what kind of evidence supports it. “Corroboration” is finite-horizon numerics and is never offered as proof.

Statement	Parameter Scope	What is Asserted	Evidence
Theorems 2 and 3	whole family	closed-form equilibria, (T, D) reduction	theorem; symbolic check
Theorem 4 and Proposition 2	whole family	Matignon classification; what the boundaries decide	theorem
Theorem 5, Corollary 1	every set with $c_T < 0$ (case (d) otherwise)	six-row atlas, α^* strictly decreasing, open in parameter space	theorem; symbolic check
Theorems 6 and 7	benchmark	rational breakpoints, explicit α^*	theorem; machine audit (Section 6)
Theorem 8, Proposition 4	any equilibrium meeting (i)–(ii)	memoryless maps invert the verdict	theorem
Corollaries 2 and 3	benchmark/every set with $c_T < 0$	explicit region of inversion	theorem
Theorem 9, Proposition 3	linear test equation, every μ	Grünwald–Letnikov schemes are verdict-faithful	theorem (Appendix A)
Section 5, Proposition 5	whole family	positivity, extinction strip, ultimate bound, global existence; what is not proved	theorem
Section 7	benchmark, Sets II–III, random ensemble	finite-horizon trajectories, convergence near α^*	corroboration only

4. Step-Size-Independent Verdict Inversion

We now make precise the memoryless-surrogate failure described in the introduction, and then place it in the dichotomy it belongs to. The object of Section 4.1 is a discrete map that discards the fractional history; it is not proposed as an integrator for Equation (1). Proper fractional schemes retain the history, and their stability theory is developed [9,10,13,15]; Section 4.2 proves what the two simplest of them do on the same problem. One terminology point: fast solvers that approximate the fractional kernel by a sum of exponentials and carry the history through an augmented state [30,31] store no explicit trajectory history and are sometimes described as memory-free. They remain approximations of the nonlocal equation and are not the object studied here; our use of “memoryless” is strict, the map $I + \varepsilon f$ containing no representation of the history whatsoever.

4.1. The Memoryless Map

Definition 2 (memoryless forward-Euler surrogate). For $\varepsilon > 0$ let $F_\varepsilon : \Omega \rightarrow \mathbb{R}^2$, $F_\varepsilon(z) = z + \varepsilon f(z)$, with f the right-hand side of Equation (1) and Ω the open half-plane Equation (2) on which f is defined. Every equilibrium of Equation (1) is a fixed point of F_ε , with multiplier spectrum $\{1 + \varepsilon\mu : \mu \in \text{spec } Df\}$. Since every coexistence equilibrium considered below lies in the interior of Ω and $F_\varepsilon(z^*) = z^*$, continuity gives, for each fixed $\varepsilon > 0$, a neighborhood of z^* on which the map and its iterates are well defined, at least until an orbit leaves that neighborhood; the instability statements below are local, and no global invariance of Ω under F_ε is asserted.

F_ε is memoryless: it uses only the current state. That is exactly what disqualifies it as an approximation of a Caputo system, and exactly what makes the following identity bite.

Lemma 2 (Euler multiplier identity). Let $\mu \in \mathbb{C}$ and $\varepsilon > 0$, and put $\lambda = 1 + \varepsilon\mu$. Then

$$|\lambda|^2 - 1 = 2\varepsilon \operatorname{Re} \mu + \varepsilon^2 |\mu|^2. \quad (20)$$

Consequently $\operatorname{Re} \mu > 0$ implies $|\lambda| > 1$ for every $\varepsilon > 0$.

Proof. Write $\mu = \rho + i\sigma$; then $|\lambda|^2 = (1 + \varepsilon\rho)^2 + \varepsilon^2\sigma^2 = 1 + 2\varepsilon\rho + \varepsilon^2(\rho^2 + \sigma^2)$, which is Equation (20). If $\rho > 0$ both terms on the right are positive for $\varepsilon > 0$. $\square$

Theorem 8 (memoryless-surrogate verdict inversion). Let z^* be an equilibrium of the commensurate Caputo system Equation (1) of order $0 < \alpha < 1$, with Jacobian $J = Df(z^*)$. Assume

- (i) every eigenvalue satisfies the sector condition: $\text{spec}(J) \subset \{\mu \neq 0 : |\arg \mu| > \alpha\pi/2\}$;
- (ii) some eigenvalue has a positive real part.

Then, z^* is locally asymptotically stable for Equation (1), while the fixed point of F_ε is linearly unstable for every $\varepsilon > 0$.

Proof. Hypothesis (i) is exactly the spectral hypothesis of the nonlinear fractional linearization theorem [18], valid for $0 < \alpha < 1$, which gives local asymptotic stability. For (ii), let $\mu \in \text{spec}(J)$ with $\operatorname{Re} \mu > 0$; then $1 + \varepsilon\mu$ is an eigenvalue of $DF_\varepsilon(z^*) = I + \varepsilon J$, and Lemma 2 gives $|1 + \varepsilon\mu| > 1$ for every $\varepsilon > 0$, so a multiplier lies outside the closed unit disc, and the fixed point is linearly unstable. The two conclusions are compatible because the sector $\{|\arg \mu| > \alpha\pi/2\}$ meets the open right half-plane whenever $\alpha < 1$. $\square$

Remark 3 (the quantifiers are not interchangeable). Hypothesis (i) is about the whole spectrum, and hypothesis (ii) is about one eigenvalue, and neither can be relaxed to the other. Weakening (i) to “some eigenvalue lies outside the sector” would be false: the Jacobian $\text{diag}(\frac{1}{48} + \frac{i}{2}, \frac{1}{10})$ at $\alpha = 9/10$ has a first eigenvalue outside the forbidden sector with positive real part, yet its second

eigenvalue is real and positive and destabilizes the equilibrium. Conversely, (ii) genuinely needs only one eigenvalue, because a single multiplier outside the unit disc suffices for linear instability of a map. A regression fixture pins in both directions.

The identity is elementary; the point is what it excludes. No choice of ε recovers the correct verdict, so the failure cannot be diagnosed or repaired by a convergence study. It is not that the surrogate is inaccurate at coarse steps and accurate at fine ones; it is qualitatively wrong at all of them.

Corollary 2 (a full region of guaranteed inversion). *For the benchmark Equation (7), let*

$$\mathcal{W} = \left\{ (A, \alpha) : \frac{2}{7} < A < \frac{10}{19}, \quad 0 < \alpha < \alpha^*(A) \right\}.$$

For every $(A, \alpha) \in \mathcal{W}$ the coexistence equilibrium E^ of Equation (1) is locally asymptotically stable, while for every $\varepsilon > 0$ the corresponding fixed point of F_ε is linearly unstable. $\mathcal{W}$ is nonempty with nonempty interior, and α^* is strictly decreasing across it from 1 to 0.*

Proof. On $2/7 < A < 10/19$, Theorem 6 gives $\Delta(A) < 0$ and $T(A) > 0$, so the spectrum is the conjugate pair $\rho \pm i\sigma$ with $\rho = T(A)/2 > 0$ and $\sigma > 0$. Both eigenvalues have the same modulus of argument, so for the $\alpha < \alpha^*(A)$ hypothesis (i) of Theorem 8 holds for the entire spectrum, by Theorem 4 (d), and hypothesis (ii) holds because $\rho > 0$. This is exactly why the corollary is not caught by Remark 3: for a conjugate pair, “some eigenvalue” and “every eigenvalue” coincide. Nonemptiness and the endpoint limits are Theorem 7. $\square$

Corollary 3 (inversion region for every parameter set with $c_T < 0$). *Let the parameter set satisfy $ea > mh$, $q < K$, and $c_T < 0$, and let $\mathcal{W}$ be the region Equation (17). For every $(A, \alpha) \in \mathcal{W}$ the coexistence equilibrium is locally asymptotically stable for Equation (1), while the fixed point of F_ε is linearly unstable for every $\varepsilon > 0$.*

Proof. On (A_T, A_2) the spectrum is a complex pair with $\operatorname{Re} \mu = T/2 > 0$ (Theorem 5 (b)), and for $\alpha < \alpha^*(A)$ both eigenvalues satisfy the sector condition (Theorem 4 (d)); Theorem 8 applies. $\square$

Figure 2 shows the mechanism at the benchmark point.

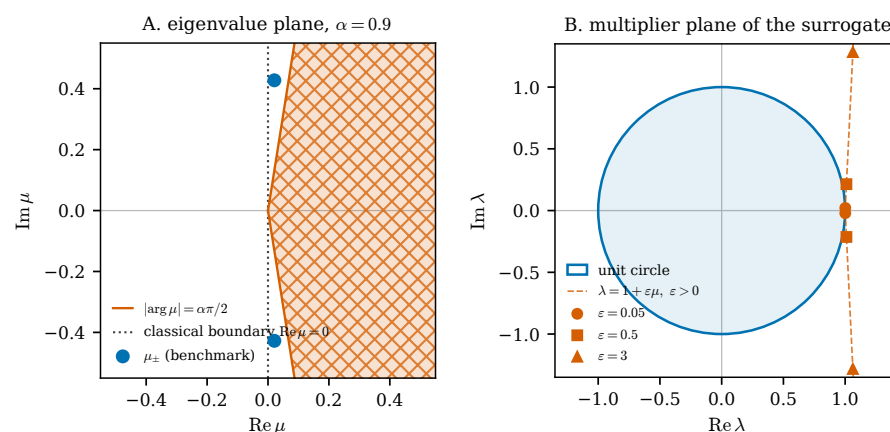


Figure 2. (A) eigenvalue plane at $\alpha = 0.9$; the hatched sector, not the right half-plane, is the Caputo-unstable set, and the benchmark pair lies outside it although $\operatorname{Re} \mu = 1/48 > 0$. (B) multiplier plane of F_ε ; by Equation (20) the image curve $\varepsilon \mapsto 1 + \varepsilon\mu$ leaves the unit disc at once and never returns. Analytic content.

4.2. History-Retaining Schemes Are Verdict-Faithful

Theorem 8 concerns the memoryless map. The natural question is what a scheme that keeps the history does on the same problem, and the answer is a dichotomy. Consider the scalar test equation

$${}^C D^\alpha y = \mu y, \quad y(0) = y_0, \quad 0 < \alpha < 1, \quad \mu \in \mathbb{C} \setminus \{0\}, \quad (21)$$

whose solution $y(t) = y_0 E_\alpha(\mu t^\alpha)$ tends to 0 iff $|\arg \mu| > \alpha\pi/2$, in which case $y(t) = -y_0(\mu \Gamma(1-\alpha) t^\alpha)^{-1}(1+o(1))$ [32]. When $\Delta \neq 0$ the linearization of each scheme below at an equilibrium of Equation (1) decouples into two copies of Equation (21) with $\mu = \mu_\pm$, so Equation (21) carries the linear stability verdict of the scheme exactly as it does for the continuous system.

Let $\omega_j = (-1)^j \binom{\alpha}{j}$, so that $\sum_{j \geq 0} \omega_j \xi^j = (1-\xi)^\alpha$, and let $u_n = y_n - y_0$. The implicit and explicit Grünwald–Letnikov schemes with step $h > 0$ are

$$h^{-\alpha} \sum_{j=0}^n \omega_j u_{n-j} = f(y_n) \quad \text{and} \quad h^{-\alpha} \sum_{j=0}^n \omega_j u_{n-j} = f(y_{n-1}), \quad n \geq 1, \quad (22)$$

the convolution quadratures of Lubich [13] generated by the backward-difference symbol. Both are first-order consistent for the Caputo problem [13,16], and both sum over the whole history at every step. Put $z = h^\alpha \mu$.

Theorem 9 (the implicit Grünwald–Letnikov scheme is verdict-faithful for every step). *Apply the implicit scheme Equation (22) to Equation (21).*

(a) If $|\arg \mu| > \alpha\pi/2$, then for every $h > 0$ the scheme is well defined, $y_n \rightarrow 0$, and

$$y_n = -\frac{y_0}{\mu \Gamma(1-\alpha) t_n^\alpha} (1+o(1)), \quad t_n = nh,$$

the same leading asymptotics as the exact solution, at every step size.

(b) If $|\arg \mu| < \alpha\pi/2$ and $0 < h < h_1(\mu, \alpha) := 2 \cos(\arg \mu/\alpha) |\mu|^{-1/\alpha}$ with $h^\alpha \mu \neq 1$, then $|y_n| \rightarrow \infty$ at an exponential rate.

In particular, on the region $\mathcal{W}$ of Corollaries 2 and 3 the implicit scheme returns the Caputo verdict at every step size, while the memoryless map returns the opposite verdict at every step size.

Proposition 3 (the explicit scheme is verdict-faithful for every sufficiently small step). *Apply the explicit scheme Equation (22) to Equation (21). If $|\arg \mu| > \alpha\pi/2$ there is $h_0(\mu, \alpha) > 0$ such that $y_n \rightarrow 0$ for every $0 < h < h_0$; if $|\arg \mu| < \alpha\pi/2$ there is $h_0 > 0$ such that $|y_n| \rightarrow \infty$ for every $0 < h < h_0$. The threshold is read off the exact boundary locus $\theta \mapsto (1 - e^{i\theta})^\alpha e^{-i\theta}$ of the scheme's stability set; at the benchmark eigenvalue and $\alpha = 9/10$ it is $h_0 = 0.505 \dots$*

Proposition 4 (every consistent memoryless one-step method inverts the verdict for small steps). *Let R be the stability function of a consistent one-step method for an ordinary differential equations, so that $R(w) = 1 + w + O(w^2)$ as $w \rightarrow 0$ [33]; applied memorylessly to Equation (1), the method has multipliers $R(\varepsilon\mu)$ at an equilibrium. If $\operatorname{Re} \mu > 0$ there is $\varepsilon_0 > 0$ such that $|R(\varepsilon\mu)| > 1$ for all $0 < \varepsilon < \varepsilon_0$. For the forward-Euler map $R(w) = 1 + w$ one may take $\varepsilon_0 = \infty$ (Lemma 2).*

The proofs are in Appendix A. The one for Theorem 9 is a generating-function argument on the closed unit disc in the spirit of [13], with the asymptotics supplied by singularity analysis [34]; Section 7.3 corroborates both statements numerically, on the test equation and on the nonlinear benchmark.

Remark 4 (what the dichotomy says, and what it does not). (i) The two families are separated by a structural property, not by accuracy. A memoryless one-step method sees the equilibrium through $R(\varepsilon\mu)$, and consistency forces $|R(\varepsilon\mu)|^2 = 1 + 2\varepsilon \operatorname{Re} \mu + O(\varepsilon^2)$: the sign of $\operatorname{Re} \mu$ decides the small-step verdict, and the method cannot know α . A history-retaining scheme sees it through the symbol $(1 - \xi)^\alpha$, whose values on the closed unit disc fill a bounded lens inside the sector $|\arg w| \leq \alpha\pi/2$ that also governs the continuous equation; the stability set of the implicit scheme is exactly the complement of that lens, and that of the explicit scheme is a bounded set tangent to the sector boundary at the origin (Section 7.3). (ii) A memoryless method can return the right verdict at isolated step sizes by coincidence: the stability region of the classical fourth-order Runge–Kutta method contains a neighborhood of the open segment $\{iy : 0 < |y| < 2\sqrt{2}\}$ of the imaginary axis and therefore enters the open right half-plane, so a ray $\arg w = \arg \mu$ very close to the imaginary axis can cross it. Proposition 4 says that no such coincidence survives mesh refinement. (iii) The Diethelm–Ford–Freed predictor–corrector method used in Section 7 also retains the history; its linear stability set is bounded and was computed by Garrappa [15]. We prove no small-step statement for it here and rely on its convergence theory [14] and on the corroboration of Section 7.3. (iv) Nothing in this section is a statement about Grünwald–Letnikov or predictor–corrector schemes being unreliable; the point is the converse, that carrying the memory is what makes a scheme able to see fractional stabilization at all.

5. Global Behavior: Uniform Ultimate Bound, Global Existence, Extinction

The results of this section are included for completeness: they are proved with standard tools (fractional first-contact arguments, the scalar Caputo comparison principle, and one linear functional), we claim no novelty of method, and their role is to place the local classification of Section 3 inside a global picture. What they do not give is stated in Proposition 5. Everything up to Corollary 4 is established on the maximal interval $[0, T_{\max})$ only.

We use one standard tool, stated for reference.

Lemma 3 (scalar Caputo comparison). *Let $0 < \alpha \leq 1$ and $c > 0$. Let g be nonnegative on $[0, T)$ and lie on every compact subinterval, in the regularity class of Proposition 1—continuous, C^1 away from 0 with $g'(t) = O(t^{\alpha-1})$, so that the Caputo derivative exists pointwise—and suppose ${}^C D^\alpha g(t) \leq -c g(t)$ on $(0, T)$. Then, $g(t) \leq g(0) E_\alpha(-c t^\alpha)$ on $[0, T)$, and in particular $g(t) \rightarrow 0$ as $t \rightarrow \infty$ when $T = \infty$. More generally, for g in the same class, if ${}^C D^\alpha g \leq -c g + \phi(t)$ with ϕ continuous, then*

$$g(t) \leq g(0) E_\alpha(-c t^\alpha) + \int_0^t (t-s)^{\alpha-1} E_{\alpha,\alpha}(-c(t-s)^\alpha) \phi(s) ds, \quad (23)$$

the kernel being nonnegative, with

$$\int_0^t \tau^{\alpha-1} E_{\alpha,\alpha}(-c \tau^\alpha) d\tau = \frac{1 - E_\alpha(-c t^\alpha)}{c} \leq \frac{1}{c}. \quad (24)$$

Proof. Both are standard; see (Chapter 7, [20]) for the comparison principle and the Mittag-Leffler resolvent representation of the scalar linear Caputo equation, and [32] for Equation (24), which follows by termwise integration of the series for $E_{\alpha,\alpha}$. $\square$

5.1. A Uniform Ultimate Bound on the Maximal Interval

Theorem 10 (uniform ultimate bound). *Let the initial data be nonnegative and let $[0, T_{\max})$ be the maximal interval of Proposition 1. Put $Q(x) = P(x) + mx$ and*

$$C = e \max_{x \geq 0} Q(x), \quad (25)$$

which is finite and nonnegative. Then, on $[0, T_{\max})$,

$$W(t) = e x(t) + y(t) \leq W(0) E_{\alpha}(-m t^{\alpha}) + \frac{C}{m} (1 - E_{\alpha}(-m t^{\alpha})) \leq \max\left\{W(0), \frac{C}{m}\right\}.$$

Proof. Q is a cubic with leading coefficient $-r/(AK) < 0$, so $Q(x) \rightarrow -\infty$ as $x \rightarrow \infty$ and the maximum over $[0, \infty)$ is attained and finite; $Q(0) = 0$ gives $C \geq 0$. By Theorem 1, $x, y \geq 0$ on $[0, T_{\max})$. Substituting $y = W - ex$ in the exact cancellation Equation (27),

$${}^C D^{\alpha} W = e P(x) - m y = -m W + e(P(x) + m x) \leq -m W + C,$$

and Lemma 3 applied to W gives the displayed estimate; the final bound follows because a convex combination of $W(0)$ and C/m never exceeds their maximum. $\square$

Corollary 4 (global existence). *For every nonnegative initial condition, $T_{\max} = \infty$: the solution of Equation (1) exists for all $t \geq 0$, and consequently*

$$\limsup_{t \rightarrow \infty} (e x(t) + y(t)) \leq \frac{C}{m},$$

a bound independent of the initial condition.

Proof. By Theorem 10 and positivity, $x \leq W/e$ and $y \leq W \leq \max\{W(0), C/m\}$ stay in a fixed compact subset of Ω on $[0, T_{\max})$, which excludes the blow-up horn of the continuation alternative for Caputo equations with locally Lipschitz right-hand side (Chapter 6, [20]), [35]; hence $T_{\max} = \infty$. The $\limsup$ follows from Theorem 10 since $E_{\alpha}(-m t^{\alpha}) \rightarrow 0$. $\square$

Corollary 5 (the invariance statements hold for all time). *For every nonnegative initial condition and every $t \geq 0$: $x(t) \geq 0$, $y(t) \geq 0$, $x(t) \leq \max\{K, x(0)\}$, and $x(t) \leq x_0$ whenever $0 < x_0 = x(0) < A$. Moreover $\limsup_{t \rightarrow \infty} (ex(t) + y(t)) \leq C/m$.*

Proof. Theorem 1 gives all four bounds on $[0, T_{\max})$, and Corollary 4 gives $T_{\max} = \infty$; the $\limsup$ is the last statement of that corollary. $\square$

Remark 5 (uniform ultimate level). *For every $\eta > 0$ and every nonnegative initial state there, is a time after which $ex(t) + y(t) \leq C/m + \eta$, and the level C/m does not involve the initial state; for the benchmark $C/m = \frac{910+103\sqrt{103}}{1350} = 1.4483\dots$ (Figure 3). We do not call the region an absorbing set, since an autonomous Caputo equation need not generate a semiflow (Remark 1); W bounds $ex + y$ and nothing sharper, and an initially over-capacity prey population is not asserted to fall below K .*

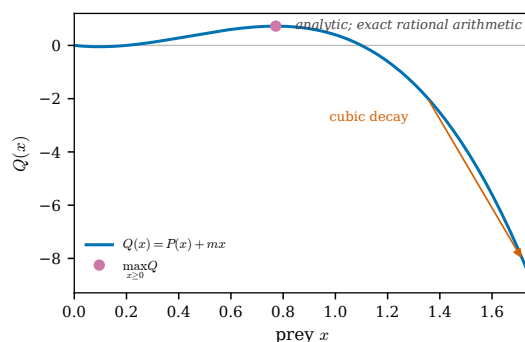


Figure 3. Why the uniform ultimate level carries no memory of the initial condition: the cubic $Q(x) = P(x) + mx$ has a finite maximum on $[0, \infty)$, attained for the benchmark at $x = \frac{13+\sqrt{103}}{30}$ with $C/m = \frac{910+103\sqrt{103}}{1350} = 1.44839\dots$, a constant of the vector field alone. Analytic content; no trajectory integrated.

5.2. The Extinction Strip

Theorem 11 (extinction of the prey). *Let $0 < A < K$, let the initial data be nonnegative, and let $x_0 := x(0)$ satisfy $0 < x_0 < A$ (for $x_0 = 0$, $x \equiv 0$ by uniqueness). Put*

$$c = r \left(1 - \frac{x_0}{K}\right) \left(1 - \frac{x_0}{A}\right) > 0. \quad (26)$$

Then, $x(t) \leq x_0 E_\alpha(-c t^\alpha) \rightarrow 0$ for every $\alpha \in (0, 1]$.

Proof. By Corollary 5 the solution satisfies $0 \leq x(t) \leq x_0$ for all $t \geq 0$, so it remains in the strip. On $[0, x_0]$ we have, writing $P(x) = -r x(1 - x/K)(1 - x/A)$,

$$P(x) = -r x \left(1 - \frac{x}{K}\right) \left(1 - \frac{x}{A}\right) \leq -c x,$$

because $1 - x/K \geq 1 - x_0/K > 0$ and $1 - x/A \geq 1 - x_0/A > 0$ there. Predation is nonpositive in the prey equation for $y \geq 0$, so ${}^C D^\alpha x \leq P(x) \leq -c x$ on the strip, and Lemma 3 gives the stated Mittag-Leffler envelope. $\square$

Lemma 4 (vanishing forcing). *Let $0 < \alpha \leq 1$, $c > 0$, and let $g \geq 0$ lie in the class of Lemma 3 on $[0, \infty)$ with ${}^C D^\alpha g \leq -c g + \phi(t)$, where ϕ is continuous, nonnegative, and $\phi(t) \rightarrow 0$. Then $g(t) \rightarrow 0$.*

Proof. By Equation (23) it suffices to treat the convolution term. Given $\eta > 0$ choose S with $\phi \leq \eta$ on $[S, \infty)$ and split at S : by nonnegativity of the kernel and Equation (24) the part over $[S, t]$ is at most η/c , while the part over $[0, S]$ is at most $\sup_{[0, S]} \phi \int_{t-S}^t \tau^{\alpha-1} E_{\alpha, \alpha}(-c \tau^\alpha) d\tau$, which tends to 0 as the tail of a convergent integral. Hence, $\limsup_{t \rightarrow \infty} g(t) \leq \eta/c$ for every $\eta > 0$. $\square$

Theorem 12 (extinction of the predator follows). *Under the hypotheses of Theorem 11, $y(t) \rightarrow 0$ as well.*

Proof. Let $W = e x + y$. The predation terms cancel exactly:

$${}^C D^\alpha W = e P(x) - m y = -m W + e P(x) + m e x. \quad (27)$$

Put $\phi(t) = |e P(x(t)) + m e x(t)|$, which is continuous and nonnegative. By Theorem 11 $x(t) \rightarrow 0$, and P is continuous with $P(0) = 0$, so $\phi(t) \rightarrow 0$. Since ${}^C D^\alpha W \leq -m W + \phi$, Lemma 4 gives $W(t) \rightarrow 0$, and as $x, y \geq 0$ with $e > 0$ this forces $y(t) \rightarrow 0$. $\square$

Proposition 5 (what is and is not proved about long-time behavior). *Let $ea > mh$ and $0 < A < q < K$.*

- (i) *The strip $\{0 \leq x_0 < A, y_0 \geq 0\}$ is contained in the basin of attraction of E_0 (Theorems 11 and 12).*
- (ii) *On the predator-free face the strip is sharp: if $y_0 = 0$ and $A < x_0 < K$, then $x(t) \geq x_0 > A$ for all $t \geq 0$, so no prey level above A leads to extinction there. Consequently, the strip cannot be widened in x_0 uniformly in y_0 .*
- (iii) *Wherever E^* is Matignon-stable, E_0 and E^* are both locally asymptotically stable: equilibrium bistability. The boundary between their basins is not characterized in this paper. With $y_0 > 0$ the extinction basin may exceed the strip, since predation can drive the prey below A , and by how much is not determined here.*
- (iv) *No global asymptotic stability statement for E^* is made, under any condition, and the set of asymptotic outcomes once E^* is Matignon-unstable is not resolved: among the equilibria classified here only E_0 is then locally asymptotically stable, and nothing more is asserted.*

Proof. (i) restates the two theorems. (ii): the face $y \equiv 0$ is invariant, by $f_2(x, 0) = 0$ and uniqueness, and on it the prey obeys ${}^C D^\alpha x = P(x)$ with $P > 0$ on (A, K) . Suppose x first reaches $x_0 - \delta$, with $0 < \delta < x_0 - A$, at some $t_* > 0$. Then, $x(t_*) < x(0)$ is the minimum of x over $[0, t_*]$, so Lemma 1 gives ${}^C D^\alpha x(t_*) < 0$, whereas $P(x(t_*)) = P(x_0 - \delta) > 0$: a contradiction. Letting $\delta \downarrow 0$ gives $x \geq x_0$ (for $\alpha = 1$ the classical first-hit argument applies). (iii) is Theorem 3 together with Theorem 4, and (iv) is a statement about what is not proved. $\square$

6. Computer-Assisted Verification

The theorems above are analytic and do not depend on the certificate layer. Independently of them, the proof objects provide audits of selected algebraic identities, benchmark constants, boundary points, and finite parameter boxes, re-derived by a separate program from the model definition and each object's recorded parameters. Each object is described below by the exact subclaim and parameter scope; its checker re-derives. A box object is uniform over the box it declares; a finite family of such objects is a regression audit of the theorem on those boxes, not a machine proof of the continuum theorem, and no such coverage is claimed. The purpose is not to replace the proofs but to let a reader check the arithmetic without trusting the authors—and the scope statement of each object is part of that contract.

What Is Exact, What Is Enclosed

The distinction that organizes the implementation is between claims that are algebraic and claims that are not.

Algebraic claims are computed in exact rational arithmetic. All parameters are exact rationals and the coexistence equilibrium has the closed-form Equation (11), so T , D and Δ are exact rationals: at the benchmark, exactly $1/24$, $11/60$, and $-2107/2880$. This matters for the critical cases. At $A = 2/7$ the trace vanishes exactly, which is what makes Theorem 4 (b) decidable and lets the checker report “stable” for $\alpha < 1$ and “critical” at $\alpha = 1$ rather than guessing. Likewise, at $A = 2/3$ the second eigenvalue of E_A vanishes exactly, which is what licenses the verdict “critical” there, and $P(K) = 0$ identically. A floating-point or interval computation could only ever report an interval containing zero, which is a strictly weaker statement: such an interval also admits nonzero values of either sign, so it cannot decide a case whose entire content is that a quantity is exactly zero.

Non-algebraic claims are enclosed with certified interval arithmetic. Interval endpoints are exact rationals; the field operations are exact rational arithmetic, with endpoints rounded

outward once they grow beyond a fixed size bound, so an enclosure can only widen. The transcendental functions actually needed— $\sqrt{\cdot}$, $\arctan$, π , and real powers—are enclosed by the Arb library, whose contract is that every returned ball provably contains the exact result.

The inventory of objects, the integrity properties enforced on them (a parameter lock and manifest completeness), the semantics of a certificate that carries an interval of parameters, and the second, independent checker are available from the corresponding author on request. Two facts about them are worth stating here. Every interval-valued object is required to contain the entire equilibrium curve over the interval it declares, so that it certifies what its parameters say; and the second checker, built on a different arithmetic library and a different formulation of the sector test, contradicts the first nowhere, returning undecided exactly on the five objects whose content is an algebraic zero that binary interval arithmetic cannot decide.

7. Numerical Corroboration

The figures and tables in this section are numerical corroboration and not proof, and each answers one theorem. Unless stated otherwise, they are produced with the Diethelm–Ford–Freed predictor–corrector method (PECE) for Caputo equations [14], which carries the full fractional memory; Section 7.3 also uses the two Grünwald–Letnikov schemes Equation (22). The memoryless map of Section 4 appears only as the object under study; it is never used to simulate Equation (1).

The solver is self-tested before use against facts it cannot fake: at $\alpha = 1$ it reproduces e^{-2} for $y' = -y$ to 1.1×10^{-8} ; for ${}^C D^\alpha y = -y$ it reproduces $E_\alpha(-1)$ to 3.7×10^{-8} at $\alpha = 0.5$ and 2.1×10^{-9} at $\alpha = 0.9$; and the observed order of convergence under successive mesh halving is 1.90, 1.90, 1.89. Every panel below states its parameters, initial condition, and mesh; machine-readable data for every plotted point ship with the figure sources.

7.1. Crossing the Critical Order

Nothing changes between the four columns of Figure 4 except α , and the qualitative behavior flips where Theorem 6 says it must.

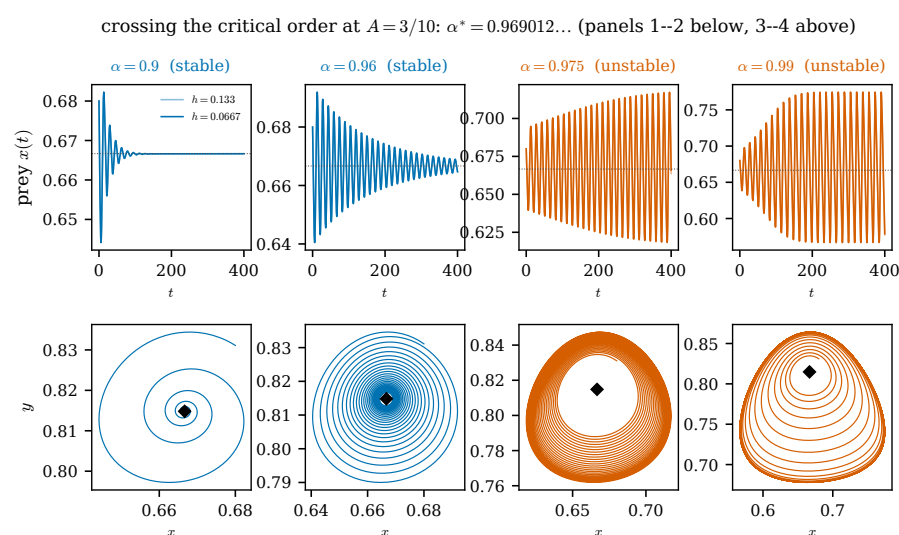


Figure 4. Crossing $\alpha^*(3/10) = 0.969012\dots$ at the benchmark from $1.02 E^*$, $T = 400$. Columns: $\alpha = 0.90, 0.96$ (Matignon-stable) and $0.975, 0.99$ (Matignon-unstable); final-to-initial distance ratios $1.1 \times 10^{-3}, 10^{-1}, 1.5, 4.3$. (Top) prey time series at $n = 3000$ and 6000 overlaid, indistinguishable. (Bottom) phase portraits.

Convergence and Horizon Sensitivity Near the Critical Order

Table 4 records what a finite-horizon simulation can and cannot decide close to α^* . For $\alpha < \alpha^*$ the approach to E^* is algebraic, like $t^{-\alpha}$; for $\alpha > \alpha^*$ the departure is exponential with the linear rate $\lambda_{\text{lin}} = |\mu|^{1/\alpha} \cos(\arg \mu/\alpha)$, which vanishes exactly at α^* . Two facts follow from the table. The mesh is not the limiting factor: the difference between the two finest meshes at $T = 400$ is below 5×10^{-4} for every α , and the observed order of convergence is 1.9. The horizon is: at $T = 400$ the endpoint diagnostic is undecided for $|\alpha - \alpha^*| \lesssim 0.006$, at $T = 800$ for $|\alpha - \alpha^*| \lesssim 0.003$, and the undecided window shrinks with T exactly as λ_{lin} predicts. The endpoint map of Section 7.4 must be read with this in mind: a cell classified “transient” near the curve $\alpha^*(A)$ is the expected finite-horizon behavior, not evidence against the theorem.

Table 4. PECE at the benchmark $A = 3/10$ from $1.02 E^*$ near $\alpha^* = 0.969012 \dots$: Matignon verdict, linear rate λ_{lin} , horizon T , finest mesh n , distance ratio $r_T = \|z(T) - E^*\|/\|z(0) - E^*\|$ at the two finest meshes, mesh-to-mesh difference, and endpoint class ($r_T < 1/2$ decaying, $r_T > 2$ growing, otherwise undecided at this horizon).

α	$\alpha - \alpha^*$	Matignon	$10^3 \lambda_{\text{lin}}$	T	n	r_T	r_T at $n/2$	Mesh Diff.	at T
0.9000	−0.0690	stable	−46.82	400	16000	0.0011	0.0011	3×10^{-10}	decaying
0.9500	−0.0190	stable	−12.87	400	16000	0.0067	0.0067	2×10^{-6}	decaying
0.9600	−0.0090	stable	−6.09	400	16000	0.099	0.099	3×10^{-5}	decaying
0.9650	−0.0040	stable	−2.71	400	16000	0.372	0.372	9×10^{-5}	decaying
0.9680	−0.0010	stable	−0.68	400	16000	0.753	0.755	2×10^{-4}	undecided
0.9690	−0.0000	stable	−0.01	400	16000	0.925	0.928	2×10^{-4}	undecided
0.9695	+0.0005	unstable	+0.33	400	16000	1.02	1.02	2×10^{-4}	undecided
0.9700	+0.0010	unstable	+0.67	400	16000	1.10	1.11	2×10^{-4}	undecided
0.9720	+0.0030	unstable	+2.02	400	16000	1.37	1.38	3×10^{-4}	undecided
0.9750	+0.0060	unstable	+4.04	400	16000	1.55	1.55	4×10^{-4}	undecided
0.9900	+0.0210	unstable	+14.13	400	16000	4.30	4.31	4×10^{-4}	growing
0.9650	−0.0040	stable	−2.71	800	32000	0.120	0.120	6×10^{-5}	decaying
0.9680	−0.0010	stable	−0.68	800	32000	0.514	0.515	2×10^{-4}	undecided
0.9690	−0.0000	stable	−0.01	800	32000	0.768	0.770	3×10^{-4}	undecided
0.9695	+0.0005	unstable	+0.33	800	32000	0.914	0.916	4×10^{-4}	undecided
0.9700	+0.0010	unstable	+0.67	800	32000	1.06	1.06	5×10^{-4}	undecided
0.9720	+0.0030	unstable	+2.02	800	32000	1.84	1.84	6×10^{-4}	undecided

7.2. Caputo Versus the Memoryless Surrogate

Figure 5 is the central contrast realised numerically: one initial condition, one parameter point, and two opposite fates depending only on whether the memory is kept. The four step sizes in panel (c) are $\varepsilon \in \{10^{-3}, 10^{-2}, 5 \cdot 10^{-2}, 10^{-1}\}$, and the exact multiplier modulus at the benchmark is $\rho_\varepsilon^2 = 1 + \varepsilon/24 + 11\varepsilon^2/60 > 1$. Shrinking ε reparametrises the same escape; it does not repair the verdict.

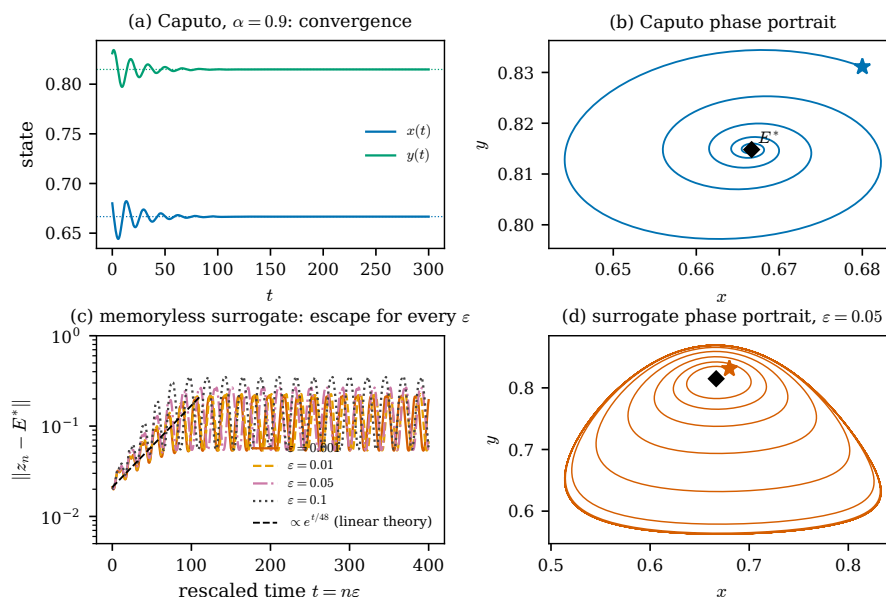


Figure 5. Theorem 8 at the benchmark $(A, \alpha) = (3/10, 9/10)$ from $1.02 E^*$. (a) Caputo time series and (b) phase portrait (PECE, $T = 300$, $n = 3000$). (c) The memoryless map escapes for every ε shown along the linear-theory slope $e^{t/48}$ in the rescaled time $t = n\varepsilon$. (d) Its orbit at $\varepsilon = 0.05$. Illustration; the theorem covers every $\varepsilon > 0$.

Other Initial Conditions

The figures above start from one perturbed initial condition. Figure 6 repeats the benchmark experiment (PECE, $T = 300$, $n = 3000$) from six initial conditions, $1.02 E^*$, $1.30 E^*$, $(0.95, 0.30)$, $(0.45, 1.40)$, $(1.20, 0.60)$ above the carrying capacity, and $(0.25, 0.60)$ inside the extinction strip $x_0 < A$. At $\alpha = 0.9$ every trajectory outside the strip approaches E^* , algebraically slowly in the tail as the Mittag-Leffler theory predicts, and the strip trajectory goes extinct as Theorem 11 requires; at $\alpha = 0.99$ the trajectories started near E^* depart from it while those started far away are still in transit at $T = 300$, about which no asymptotic claim is made.

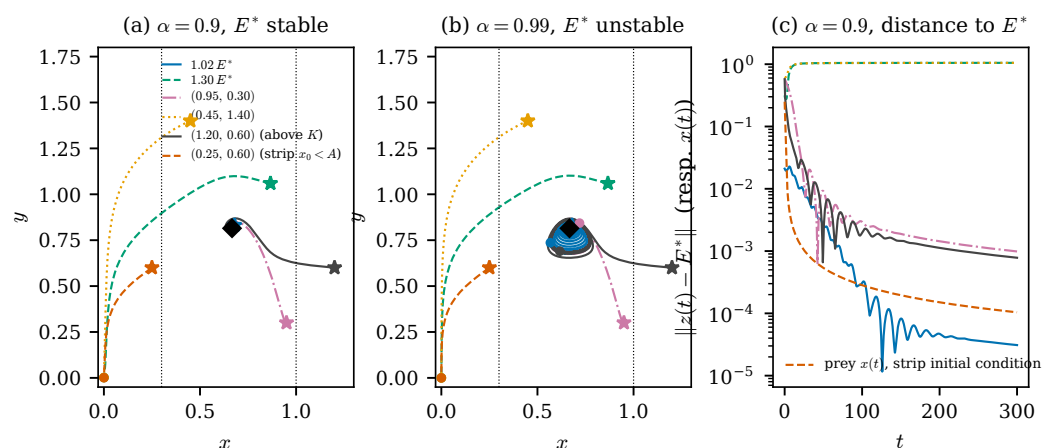


Figure 6. Six initial conditions at the benchmark $A = 3/10$, listed in the legend of (a), at (a) $\alpha = 0.9$ and (b) $\alpha = 0.99$: stars mark initial states, dots the states at T , the diamond E^* , dotted lines $x = A$ and $x = K$. (c) Distance to E^* at $\alpha = 0.9$, and the prey biomass of the strip trajectory.

7.3. History-Retaining Schemes on the Benchmark

Figure 7 corroborates Theorem 9 and Propositions 3 and 4 at the benchmark eigenvalue $\mu = \frac{1}{48} + \frac{i}{2}\sqrt{2107/2880}$ and $\alpha = 0.9$, for which $|\arg \mu| = 1.5221 > \alpha\pi/2 = 1.4137$ and $\operatorname{Re} \mu > 0$. Panel (a) shows the sets described in Remark 4 (i) in one plane; the ray in the

direction of $\arg \mu$ is the set of all scaled eigenvalues over all step sizes. Panel (b) shows the implicit scheme on the test equation at $h = 0.1, 1, 10$, and 100 : all four runs follow the exact solution and the algebraic asymptote of Theorem 9 (a), and the normalized tail $y_n \mu \Gamma(1 - \alpha) t_n^\alpha$ equals -1.000 to three decimals at the end of every run. Panel (c) is the verdict as a function of the step: the memoryless map amplifies for every ε , the implicit scheme damps for every h , and the explicit scheme damps exactly for $h < h_0 = 0.505$. Panel (d) is the nonlinear benchmark from $1.02 E^*$: PECE and both Grünwald–Letnikov schemes are within 3×10^{-5} of E^* at $T = 300$, at step sizes from 0.05 to 1 , while the memoryless orbits saturate at a distance of order 10^{-1} .

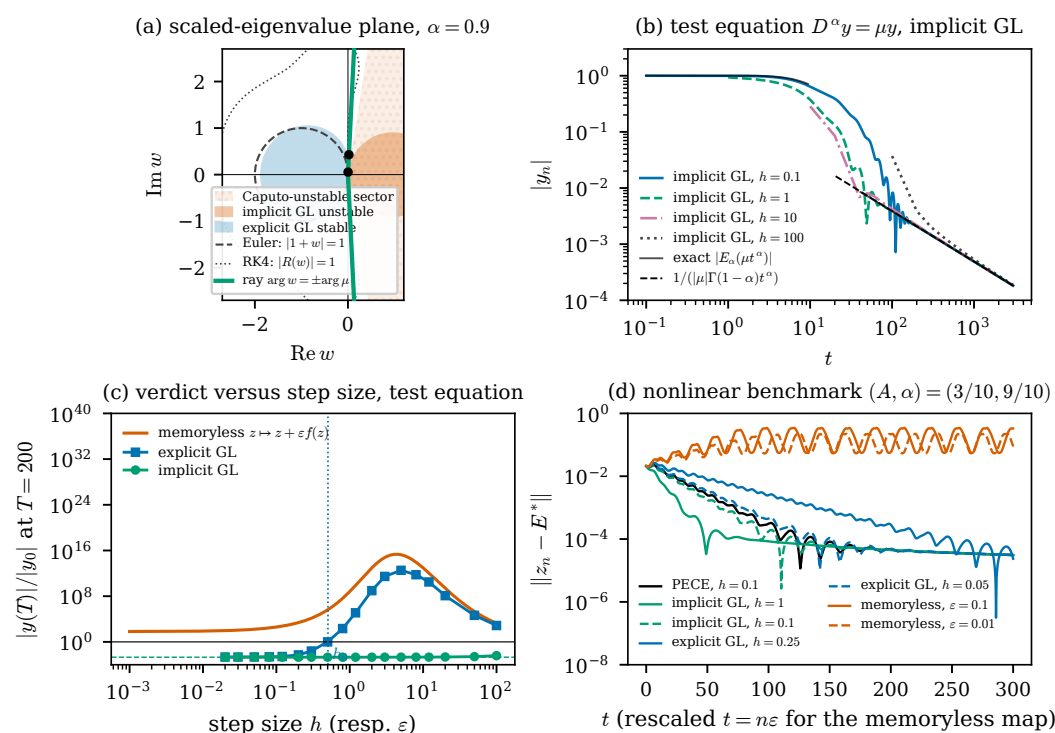


Figure 7. The dichotomy of Section 4.2 at the benchmark eigenvalue, $\alpha = 0.9$. (a) Scaled-eigenvalue plane: the stability sets of Remark 4 (i) and the ray of scaled benchmark eigenvalues. (b) Implicit scheme on the test equation at four step sizes, with the exact solution and asymptote. (c) Amplitude at $T = 200$ against the step size. (d) Nonlinear benchmark. (a) analytic; (b)–(d) corroboration.

7.4. Continuation and Regime Maps

Figure 8 gives the one-parameter view, the branch with its exact breakpoints and the displacement of the stability switch by the fractional order, and Figure 9 gives the two-parameter view: finite-horizon endpoints laid under the exact atlas boundary, never asymptotic claims. Each cell of the map is classified literally at $T = 200$: near E^* if $\|z(T) - E^*\| < 10^{-2}$, low prey if $x(T) < 10^{-3}$, otherwise transient. The extinction-strip theorem does not apply to these runs, whose initial prey level $1.05 x^* = 0.7$ exceeds every A shown.

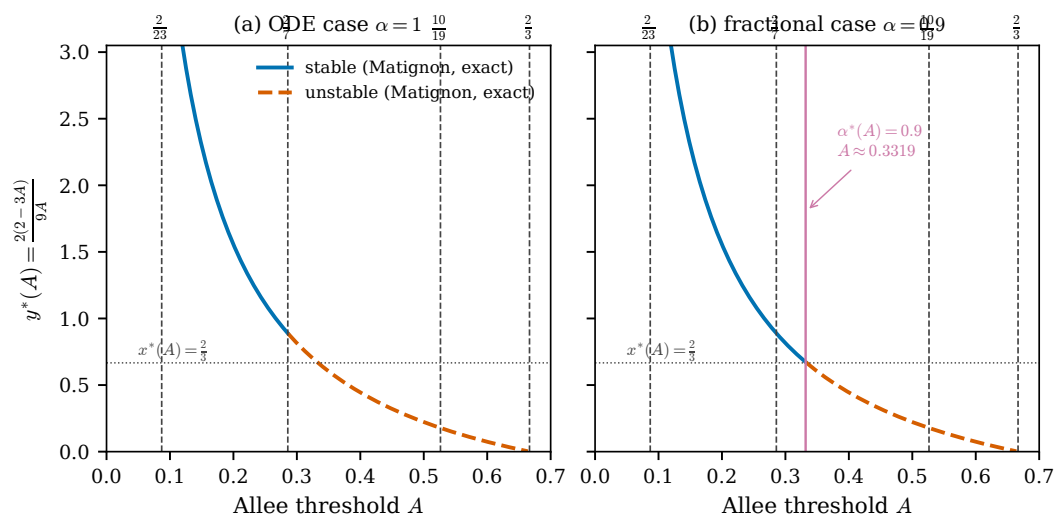


Figure 8. The coexistence branch $y^*(A) = 2(2 - 3A)/(9A)$, $x^* = 2/3$, with exact stability coding from the rational $T(A)$, $D(A)$: solid Matignon-stable, dashed Matignon-unstable. **(a)** $\alpha = 1$: the switch is at $A = 2/7$. **(b)** $\alpha = 0.9$: it moves to the solution of $\alpha^*(A) = 0.9$, $A \approx 0.3319$. Analytic content.

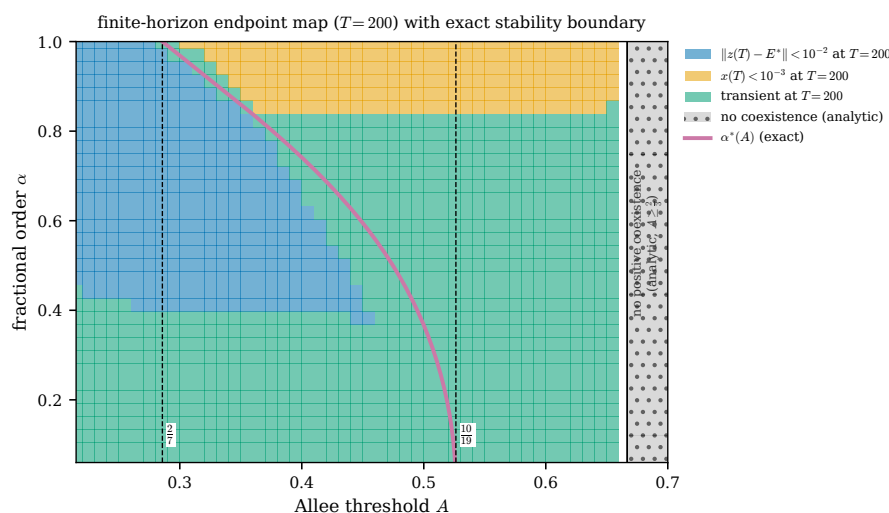


Figure 9. Finite-horizon endpoint map: one PECE run per cell from $1.05E^*(A)$, $T = 200$, $n = 1600$, classified at T as near E^* (blue), low prey (orange), or transient (green); grey: no coexistence equilibrium. The curve $\alpha^*(A)$ and the breakpoints are exact. Transient cells near the curve are the horizon effect of Table 4. Finite-horizon diagnostics; exact overlay.

7.5. Beyond the Benchmark

Theorem 5 and Corollary 1 say that the benchmark atlas is one instance of an open phenomenon. Figure 10 and Table 5 exhibits it on two further parameter sets chosen to have no rational structure: Set II, with generic rational coefficients $r = 1$, $a = 5/2$, $h = 3/2$, $e = 11/20$, $m = 1/2$ ($q = 4/5$), and Set III, with irrational coefficients $r = \sqrt{2}$, $a = 1 + \sqrt{3}$, $h = \pi/4$, $e = 1/\sqrt{3}$, $m = 1/\sqrt{2}$ ($q = 0.6919\dots$), both with $K = 1$. In both, $c_T < 0$ and the six-row atlas appears with its own breakpoints; for Set III every breakpoint is a certified Arb enclosure of width below 10^{-6} , computed from Equation (15) with the same arithmetic as the proof objects. Inside the wedge of Set II, at $(A, \alpha) = (0.62, 0.75)$ where $\alpha^*(A) = 0.839$ and $\text{Re } \mu = 0.0287 > 0$, PECE ($h = 0.1$) and the implicit Grünwald–Letnikov scheme ($h = 0.5$) converge to E^* while the memoryless map escapes at $\varepsilon = 0.1$ and 0.01 (Figure 10c): the dichotomy is not a property of the benchmark.

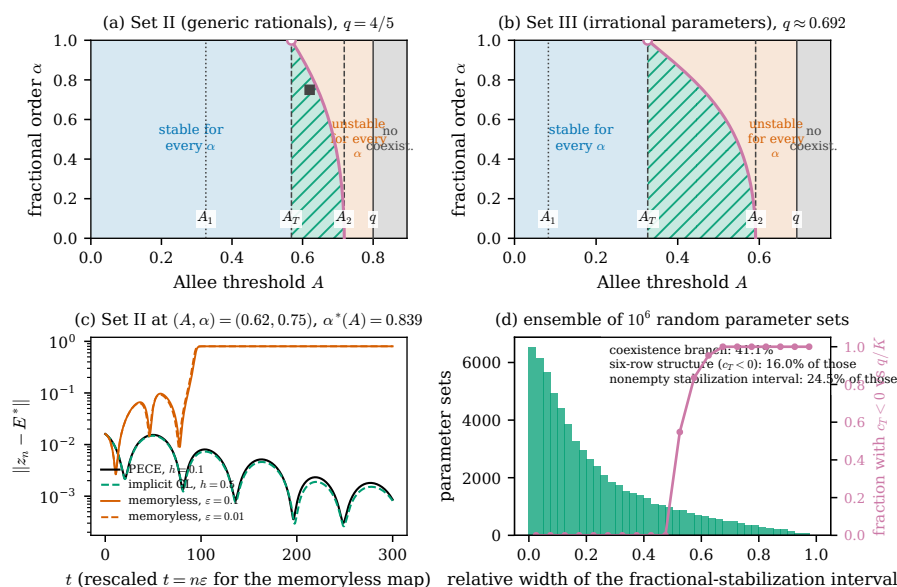


Figure 10. The atlas beyond the benchmark. (a) Set II and (b) Set III, breakpoints in Table 5; the square in (a) is the point of (c). (c) Set II inside its wedge: PECE, implicit scheme, memoryless map. (d) Ensemble of 10^6 random parameter sets: width of the stabilization interval, and fraction with $c_T < 0$ against q/K . (a), (b) analytic; (c,d) corroboration.

How common is the structure? In an ensemble of 10^6 parameter sets drawn log-uniformly from $r, a \in [0.1, 10]$, $h \in [0.01, 10]$, $e \in [0.05, 1]$, $m \in [0.01, 5]$ with $K = 1$, a coexistence branch ($ea > mh$, $q < K$) exists for 41.1% of the sets; of these, 16.0% satisfy $c_T < 0$ and therefore carry the full six-row atlas with α^* ranging over all of $(0, 1)$, and 24.5% have a nonempty fractional-stabilization interval (the difference being sets with $c_T \geq 0$, where $\alpha^* < 1$ on the interval, Theorem 5 (d)). Among the six-row sets the relative width $(A_2 - A_T)/q$ of the stabilization interval has median 0.16 and interquartile range $[0.07, 0.34]$, and the fraction with $c_T < 0$ rises steeply with q/K and reaches one for $q/K \geq 2/3$, as Corollary 1 predicts. These are statistics over an arbitrary sampling measure and carry no ecological weight; their content is that the phenomenon occupies a set of parameters with positive measure, not a fine-tuned point.

Table 5. Breakpoints of the atlas for the three parameter sets, from Equation (15). Exact rationals where the data are rational; otherwise Arb enclosures, printed with outward rounding to six decimals.

	Set I (benchmark)	Set II	Set III
q	2/3	4/5	[0.691892, 0.691893]
c_T	-1/4	-108/275	[-0.269381, -0.269380]
A_1 (zero of Δ)	2/23	[0.325837, 0.325838]	[0.083007, 0.083008]
A_T (zero of T)	2/7	54/95	[0.326494, 0.326495]
A_2 (zero of Δ)	10/19	[0.718094, 0.718095]	[0.590981, 0.590982]
arithmetic	exact rational	exact rational; Arb for the zeros of Δ	Arb enclosures throughout

8. Discussion and Conclusions

8.1. What Is New, and What Is for Completeness

The new mathematics is in Sections 3.3 and 4: the affine structure of (T, D) in $1/A$, which turns the benchmark atlas into an instance of a theorem valid on an open set of parameters, and the discretization dichotomy, which separates memoryless one-step maps from history-retaining schemes by a structural property on an exactly described region. The reduction in Section 3 is what both rest on. The global results of Section 5 are standard in method and included for completeness, and Proposition 5 states what they leave open.

The proof objects and the numerics are evidence about constants and trajectories, not part of any proof.

8.2. What the Dichotomy Means for Method Choice

On the fractional-stabilization region, an explicit memoryless map does not merely lose accuracy; it returns the opposite qualitative answer at every step size, and so does every consistent memoryless one-step method at all sufficiently small steps. The remedy is not a smaller step but a scheme that carries the history: the implicit Grünwald–Letnikov scheme returns the correct verdict at every step size and reproduces the exact algebraic tail, the explicit one at every sufficiently small step, and the predictor–corrector method in practice. Near the critical order, every method is slow, for the reason quantified in Table 4, and a finite-horizon classification there is undecided by nature rather than by defect.

8.3. Which Conclusions Admit an Ecological Reading, and Which Do Not

The paper is mathematical, and the benchmark is not calibrated (Definition 1); the rational breakpoints carry no ecological meaning. Four statements do have a reading in the model's own terms. (i) The Allee threshold A is a genuine extinction threshold for the prey: below it, the prey goes extinct with a Mittag-Leffler envelope whatever the predator does, and on the predator-free face the threshold is sharp (Proposition 5). (ii) Coexistence requires $A < q < K$: the prey level q at which the predator breaks even must lie between the Allee threshold and the carrying capacity. (iii) Fractional stabilization arises under $c_T < 0$, which holds whenever $q/K \geq 2/3$ and for small handling time reduces to $q/K > 1/2$ (Corollary 1): a predator whose break-even prey level is high relative to the carrying capacity is the structural setting in which memory can stabilize coexistence that classical linearization rejects. (iv) The order α is read here, as in the modeling literature [2,36,37], as a phenomenological summary of history dependence of past densities acting on present demographic rates through delayed or cumulative effects, and not as a measured quantity. Whether a given population has $\alpha < 1$ in this sense, and which value is an empirical question we do not address. What the atlas shows is that the qualitative outcome depends on α discontinuously at $\alpha^*(A)$, so a modeling decision about memory is not a matter of fine-tuning. An applied reading of any specific number in this paper would require fitting r, K, a, h, e, m and α to data, which we have not done.

8.4. Evidence Boundaries

Four categories are kept apart throughout, because blurring them is the failure mode of computer-assisted work. Theorem: Sections 3–5 and Appendix A. Certified computation: the proof objects of Section 6 and the enclosures of Table 5, exactly where the claim is algebraic and enclosed otherwise. Numerical corroboration: Section 7. Open: the items below. A theorem may also carry a partial machine audit of selected constants; such an audit is additional evidence and does not change the analytic status of the theorem.

8.5. Open Problems

The boundary between the basins of E_0 and E^* , and any global asymptotic stability statement for E^* , are not addressed (Proposition 5). Nothing is asserted about the nonlinear dynamics at the critical boundaries of Proposition 2. A small-step verdict-faithfulness statement for the predictor–corrector method, analogous to Proposition 3, is not proved here. Certification over a parameter box that straddles an algebraic or Matignon boundary returns undecided by design; a compact wide-box certificate returning a verdict with an explicit boundary decomposition remains to be built. Periodic orbits, chaotic dynamics, and control of Equation (1) are outside the scope of this paper.

8.6. Reproducibility

The LaTeX source, the figure and table generation scripts with their machine-readable output data, the proof objects with a completeness-checked SHA-256 manifest, the two checkers, the symbolic verification script, and a test suite with negative fixtures are available from the corresponding author on request; two commands re-derive every object and rebuild every figure, table, and the paper.

8.7. Conclusions

For a strong-Allee, Holling type-II predator–prey system under a commensurate Caputo derivative, the local stability of the coexistence equilibrium reduces exactly to two scalars that are affine in the reciprocal Allee threshold. Under one explicit open inequality every parameter set carries the same six-row stability atlas with a strictly decreasing critical order, and the benchmark atlas with rational breakpoints is one instance of it. The atlas contains a region where memory stabilizes an equilibrium that classical linearization rejects, and on that region discretization splits along a structural line: memoryless one-step maps invert the verdict for every step size in the Euler case and for all small steps in general, while the Grünwald–Letnikov schemes, which retain the history, return it, for every step size in the implicit case and for all small steps in the explicit one. Positivity, a sharp-on-face extinction strip, a uniform ultimate bound, and global existence complete the picture; selected constants are audited by machine-checkable objects, and the numerical evidence extends to two further parameter sets and to a random ensemble.

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Appendix A Proofs for Section 4.2

Throughout, $(1 - \zeta)^\beta := \exp(\beta \operatorname{Log}(1 - \zeta))$ with the principal logarithm. It is analytic on $\mathbb{C} \setminus [1, \infty)$, hence, on the open unit disc $\mathbb{D}$, and continuous on $\overline{\mathbb{D}} \setminus \{1\}$. For $\zeta \in \overline{\mathbb{D}} \setminus \{1\}$ the point $1 - \zeta$ lies in the closed disc $|w - 1| \leq 1$ punctured at 0, so $|\operatorname{Arg}(1 - \zeta)| < \pi/2$ and

$$(1 - \zeta)^\alpha \in S_\alpha := \{w \neq 0 : |\operatorname{Arg} w| < \alpha\pi/2\}, \quad |(1 - \zeta)^\alpha| \leq 2^\alpha. \quad (\text{A1})$$

We use two facts from singularity analysis [34, Thms. VI.1 and VI.3]. A Δ -domain at 1 is a set $\{\zeta : |\zeta| < 1 + \eta, \zeta \neq 1, |\operatorname{Arg}(\zeta - 1)| > \varphi\}$ with $\eta > 0$ and $0 < \varphi < \pi/2$. If $G(\zeta) = \sum g_n \zeta^n$ is analytic in a Δ -domain and $G(\zeta) = c(1 - \zeta)^{-\beta} + O(|1 - \zeta|^{-\gamma})$ as $\zeta \rightarrow 1$ there, with $\beta \notin \{0, -1, -2, \dots\}$ and $\gamma < \beta$, then

$$g_n = \frac{c}{\Gamma(\beta)} n^{\beta-1} + O(n^{\gamma-1}). \quad (\text{A2})$$

Proof of Theorem 9. Let $u_n = y_n - y_0$, so $u_0 = 0$, and write the scheme for $n \geq 1$ as $\sum_{j=0}^n \omega_j u_{n-j} = z y_n = z(u_n + y_0)$ with $z = h^\alpha \mu$. The coefficient of u_n is $\omega_0 - z = 1 - z$, so the scheme is well defined iff $z \neq 1$; under either hypothesis z is not a positive real number in case (a), and $z \neq 1$ is assumed in case (b). Let $U(\zeta) = \sum_{n \geq 0} u_n \zeta^n$ and

$Y(\xi) = \sum_{n \geq 0} y_n \xi^n = U(\xi) + y_0/(1 - \xi)$ as formal power series. Multiplying the scheme by ξ^n and summing over $n \geq 1$ (the $n = 0$ term of the convolution is $\omega_0 u_0 = 0$, so the sum may start at $n = 0$ on the left) gives $(1 - \xi)^\alpha U(\xi) = z(Y(\xi) - y_0) = zU(\xi) + zy_0\xi/(1 - \xi)$, hence

$$Y(\xi) = y_0 \frac{(1 - \xi)^{\alpha-1} - z}{(1 - \xi)^\alpha - z} =: y_0 \frac{N(\xi)}{g(\xi)}. \quad (\text{A3})$$

The identity holds as a formal power series, and both sides are analytic wherever $g \neq 0$ in $\mathbb{D}$, so the coefficients of the analytic function on the right are the y_n .

(a) Let $|\arg \mu| > \alpha\pi/2$, so $z \notin \overline{S}_\alpha$. By Equation (A1), $g(\xi) = (1 - \xi)^\alpha - z \neq 0$ on $\overline{\mathbb{D}} \setminus \{1\}$, and $g(1) = -z \neq 0$. Since $(1 - \xi)^\alpha \rightarrow 0$ as $\xi \rightarrow 1$ in $\mathbb{C} \setminus [1, \infty)$, g is bounded away from zero on a neighborhood of 1 in that slit plane; on the rest of $\overline{\mathbb{D}}$, a compact set on which g is continuous and nonvanishing, g is bounded away from zero as well, and by uniform continuity the same holds on $\{|\xi| \leq 1 + \eta\} \setminus [1, \infty)$ for some $\eta > 0$. Hence, Y is analytic in a Δ -domain. As $\xi \rightarrow 1$, $1/g(\xi) = -z^{-1}(1 - (1 - \xi)^\alpha/z)^{-1} = -z^{-1}(1 + O(|1 - \xi|^\alpha))$, so

$$\begin{aligned} Y(\xi) &= -\frac{y_0}{z} (1 - \xi)^{\alpha-1} + O(|1 - \xi|^{2\alpha-1}) + O(1) \\ &= -\frac{y_0}{z} (1 - \xi)^{-(1-\alpha)} + O(|1 - \xi|^{-\gamma}), \quad \gamma = \max\{1 - 2\alpha, 0\} < 1 - \alpha. \end{aligned}$$

By Equation (A2) with $\beta = 1 - \alpha \in (0, 1)$,

$$y_n = -\frac{y_0}{z\Gamma(1-\alpha)} n^{-\alpha} + O(n^{\gamma-1}), \quad \gamma - 1 < -\alpha,$$

and $z n^\alpha = \mu (nh)^\alpha = \mu t_n^\alpha$ gives the stated asymptotics; in particular $y_n \rightarrow 0$. The exact solution has the same leading term [32, Ch. 1].

(b) Let $|\arg \mu| < \alpha\pi/2$ and $0 < h < h_1$. Then $|\arg z| = |\arg \mu| < \alpha\pi/2$, so $z^{1/\alpha} := \exp(\text{Log } z/\alpha)$ satisfies $|\arg z^{1/\alpha}| < \pi/2$ and $(z^{1/\alpha})^\alpha = z$. Put $\xi^* = 1 - z^{1/\alpha}$. Then, $|\xi^*| < 1$ iff $|z|^{2/\alpha} < 2|z|^{1/\alpha} \cos(\arg \mu/\alpha)$, i.e., iff $h|\mu|^{1/\alpha} < 2 \cos(\arg \mu/\alpha)$, which is $h < h_1$. The map $w \mapsto w^\alpha$ is injective on the right half-plane, so ξ^* is the only zero of g in $\overline{\mathbb{D}}$, and it is simple because $g'(\xi^*) = -\alpha(1 - \xi^*)^{\alpha-1} \neq 0$. The numerator at the pole is $N(\xi^*) = z^{(\alpha-1)/\alpha} - z = z(z^{-1/\alpha} - 1)$, which vanishes only if $z^{-1/\alpha} = 1$, i.e., $z = 1$; this is excluded. Hence $Y = c/(1 - \xi/\xi^*) + \tilde{Y}$ with $c \neq 0$ and $\tilde{Y}$ analytic in $\mathbb{D}$, with the same Δ -domain analyticity and the same expansion at 1 as in (a); so $y_n = c(\xi^*)^{-n} + O(n^{-\alpha})$ and $|y_n| \rightarrow \infty$ like $|\xi^*|^{-n}$.

The last sentence of the theorem is (a) applied to $\mu_\pm$, whose arguments exceed $\alpha\pi/2$ on $\mathcal{W}$, together with Theorem 8. $\square$

Proof of Proposition 3. With u_n , U , Y , and z as above, the explicit scheme reads $\sum_{j=0}^n \omega_j u_{n-j} = z y_{n-1}$ for $n \geq 1$, and summing gives $(1 - \xi)^\alpha U(\xi) = z\xi Y(\xi) = z\xi U(\xi) + z\xi y_0/(1 - \xi)$, hence

$$Y(\xi) = y_0 \frac{(1 - \xi)^{\alpha-1}}{g_e(\xi)}, \quad g_e(\xi) := (1 - \xi)^\alpha - z\xi. \quad (\text{A4})$$

The scheme is always well defined ($\omega_0 = 1$). Its stability set is $\{z : g_e \text{ has no zero in } \overline{\mathbb{D}}\}$, and $g_e(e^{i\theta}) = 0$ iff $z = (1 - e^{i\theta})^\alpha e^{-i\theta}$, which is the boundary locus in the statement.

Localization. If $\xi \in \overline{\mathbb{D}}$ and $g_e(\xi) = 0$ then $|1 - \xi|^\alpha = |\xi| |\xi| \leq |z|$, so $|1 - \xi| \leq |z|^{1/\alpha}$: for small $|z|$ every zero in $\overline{\mathbb{D}}$ lies near 1; once $|1 - \xi| \leq 1/2$ we have $|\xi| \geq 1/2$ and $|\arg \xi| \leq \frac{\pi}{2} |\sin \arg \xi| = \frac{\pi}{2} |\text{Im } \xi|/|\xi| \leq \pi|1 - \xi| \leq \delta(z) := \pi|z|^{1/\alpha} \rightarrow 0$. Writing $1 - \xi = \rho e^{i\varphi}$ with

$|\varphi| < \pi/2$ and taking arguments in $(1 - \xi)^\alpha = z\xi$ gives $\alpha\varphi \equiv \text{Arg } z + \text{Arg } \xi \pmod{2\pi}$. Both $|\alpha\varphi| < \alpha\pi/2 < \pi/2$ and $|\text{Arg } \xi| \leq \delta$, so for $\delta < \pi/2$ the congruence is an equality and

$$|\text{Arg } z| \leq \alpha\pi/2 + \delta(z). \quad (\text{A5})$$

Case $|\arg \mu| > \alpha\pi/2$. Since $\text{Arg } z = \arg \mu$, choose h_0 so that $\delta(h^\alpha \mu) < |\arg \mu| - \alpha\pi/2$ for $h < h_0$. Then Equation (A5) fails for every $\xi \in \mathbb{D}$, so g_e has no zero there; $g_e(1) = -z \neq 0$, and as in part (a) above, Y is analytic in a Δ -domain with $1/g_e(\xi) = -z^{-1}(1 + O(|1 - \xi|^\alpha))$ as $\xi \rightarrow 1$. Hence, $Y = -(y_0/z)(1 - \xi)^{\alpha-1} + O(|1 - \xi|^{2\alpha-1})$, and Equation (A2) gives $y_n = -y_0(\mu\Gamma(1 - \alpha)t_n^\alpha)^{-1}(1 + o(1)) \rightarrow 0$: the explicit scheme, too, reproduces the exact tail whenever it is stable.

Case $|\arg \mu| < \alpha\pi/2$. Let $\xi_0 = 1 - z^{1/\alpha}$, the zero of $g(\xi) = (1 - \xi)^\alpha - z$, and put $\kappa = \frac{1}{2} \cos(\arg \mu/\alpha) \in (0, \frac{1}{2}]$ and $\Gamma_z = \{\xi : |\xi - \xi_0| = \kappa|z|^{1/\alpha}\}$. On Γ_z , $1 - \xi = z^{1/\alpha}(1 + \kappa e^{i\theta})$, so $g(\xi) = z[(1 + \kappa e^{i\theta})^\alpha - 1]$ and $|g| \geq c(\alpha, \kappa)|z|$ with $c(\alpha, \kappa) = \min_\theta |(1 + \kappa e^{i\theta})^\alpha - 1| > 0$ (the minimum is positive because $(1 + \kappa e^{i\theta})^\alpha = 1$ forces $1 + \kappa e^{i\theta} = 1$). Also $|g_e - g| = |z||1 - \xi| \leq (1 + \kappa)|z|^{1+1/\alpha}$ on Γ_z , which is smaller than $c(\alpha, \kappa)|z|$ once $|z|^{1/\alpha} < c/(1 + \kappa)$. By Rouché's theorem, g_e has exactly one zero ξ_e inside Γ_z . The circle lies in $\mathbb{D}$: with $w = 1 - \xi$, $|\xi| < 1$ iff $|w|^2 < 2 \text{Re } w$, and on Γ_z , $\text{Re } w \geq |z|^{1/\alpha}(\cos(\arg \mu/\alpha) - \kappa) = \kappa|z|^{1/\alpha}$ while $|w|^2 \leq (1 + \kappa)^2|z|^{2/\alpha}$, so the inequality holds once $|z|^{1/\alpha} < 2\kappa/(1 + \kappa)^2$. The zero is simple for small $|z|$, because $g'_e(\xi) = -\alpha(1 - \xi)^{\alpha-1} - z$ and $|\alpha(1 - \xi)^{\alpha-1}| \asymp |z|^{(\alpha-1)/\alpha} \rightarrow \infty$ while $|z| \rightarrow 0$. The numerator $(1 - \xi)^{\alpha-1}$ in Equation (A4) does not vanish, so Y has a simple pole with nonzero residue at $\xi_e \in \mathbb{D}$ and $|y_n| \rightarrow \infty$ like $|\xi_e|^{-n}$. Taking h_0 so that $|z| = h^\alpha|\mu|$ satisfies the two smallness conditions for $h < h_0$ completes the proof. $\square$

Proof of Proposition 4. $|R(\varepsilon\mu)|^2 = |1 + \varepsilon\mu + O(\varepsilon^2)|^2 = 1 + 2\varepsilon \text{Re } \mu + O(\varepsilon^2)$, which exceeds 1 for all sufficiently small $\varepsilon > 0$ when $\text{Re } \mu > 0$. For $R(w) = 1 + w$ the $O(\varepsilon^2)$ term is $\varepsilon^2|\mu|^2 \geq 0$, which is Lemma 2. $\square$

References

- Matignon, D. Stability results for fractional differential equations with applications to control processing. In Proceedings of the Computational Engineering in Systems Applications, Lille, France, 9–12 July 1996; pp. 963–968.
- Ahmed, E.; El-Sayed, A.M.A.; El-Saka, H.A.A. Equilibrium points, stability and numerical solutions of fractional-order predator–prey and rabies models. *J. Math. Anal. Appl.* **2007**, *325*, 542–553. <https://doi.org/10.1016/j.jmaa.2006.01.087>.
- Saha, A.; Pal, D.; Kesh, D.; Mukherjee, D. A fractional-order commensurate and incommensurate study on predator–prey system with Allee in predator and fear effect. *Fract. Calc. Appl. Anal.* **2026**, *29*, 1486–1515. <https://doi.org/10.1007/s13540-026-00515-8>.
- Ramesh, K.; Ranjith Kumar, G.; Khan, A.; Abdeljawad, T. Analysis of the stability of a predator–prey model including the memory effect, double Allee effect and Holling type-I functional response. *PLoS ONE* **2025**, *20*, e0305179. <https://doi.org/10.1371/journal.pone.0305179>.
- Wang, Q.; Han, R. Dynamics of fractional order predator prey models with refuge and Allee effect in prey: With and without time delay. *Qual. Theory Dyn. Syst.* **2025**, *24*, 49. <https://doi.org/10.1007/s12346-024-01212-8>.
- Baghel, R.S. Hopf bifurcation and stability in a fractional-order age-structured predator–prey model with Allee effect and dual delays. *Chaos Solitons Fractals* **2026**, *205*, 117880. <https://doi.org/10.1016/j.chaos.2026.117880>.
- Mondal, C.; Mondal, R.; Kesh, D.; Mukherjee, D. Dynamics of predator–prey system with the consequences of double Allee effect in prey population. *J. Biol. Phys.* **2025**, *51*, 5. <https://doi.org/10.1007/s10867-025-09670-0>.
- Rahmi, E.; Anggriani, N.; Panigoro, H.S.; Peter, O.J. A fractional-order approach to predator–prey interactions: Modeling fear and disease dynamics with memory effects. *PLoS ONE* **2026**, *21*, e0339351. <https://doi.org/10.1371/journal.pone.0339351>.
- Čermák, J.; Kisela, T.; Nechvátal, L. Stability regions for linear fractional differential systems and their discretizations. *Appl. Math. Comput.* **2013**, *219*, 7012–7022. <https://doi.org/10.1016/j.amc.2012.12.019>.
- Cresson, J.; Szafrńska, A. Discrete and continuous fractional persistence problems—The positivity property and applications. *Commun. Nonlinear Sci. Numer. Simul.* **2017**, *44*, 424–448. <https://doi.org/10.1016/j.cnsns.2016.07.016>.
- Elsadany, A.A.; Matouk, A.E. Dynamical behaviors of fractional-order Lotka–Volterra predator–prey model and its discretization. *J. Appl. Math. Comput.* **2015**, *49*, 269–283. <https://doi.org/10.1007/s12190-014-0838-6>.
- Mondal, S.; Biswas, M.; Bairagi, N. Local and global dynamics of a fractional-order predator–prey system with habitat complexity and the corresponding discretized fractional-order system. *J. Appl. Math. Comput.* **2020**, *63*, 311–340. <https://doi.org/10.1007/s12190-020-01319-6>.

13. Lubich, C. Discretized fractional calculus. *SIAM J. Math. Anal.* **1986**, *17*, 704–719. <https://doi.org/10.1137/0517050>.
14. Diethelm, K.; Ford, N.J.; Freed, A.D. A predictor-corrector approach for the numerical solution of fractional differential equations. *Nonlinear Dynam* **2002**, *29*, 3–22.
15. Garrappa, R. On linear stability of predictor–corrector algorithms for fractional differential equations. *Int. J. Comput. Math.* **2010**, *87*, 2281–2290. <https://doi.org/10.1080/00207160802624331>.
16. Garrappa, R. Numerical solution of fractional differential equations: A survey and a software tutorial. *Mathematics* **2018**, *6*, 16. <https://doi.org/10.3390/math6020016>.
17. Rauh, A.; Jaulin, L. Novel techniques for a verified simulation of fractional-order differential equations. *Fractal Fract.* **2021**, *5*, 17. <https://doi.org/10.3390/fractalfract5010017>.
18. Cong, N.D.; Doan, T.S.; Siegmund, S.; Tuan, H.T. Linearized asymptotic stability for fractional differential equations. *arXiv* **2016**, arXiv:1512.04989.
19. Cong, N.D.; Doan, T.S.; Siegmund, S.; Tuan, H.T. An instability theorem for nonlinear fractional differential systems. *Discret. Contin. Dyn. Syst. Ser. B* **2017**, *22*, 3079–3090.
20. Diethelm, K. The Analysis of Fractional Differential Equations. In *Lecture Notes in Mathematics 2004*; Springer: Berlin/Heidelberg, Germany, 2010.
21. Lubich, C. Runge–Kutta theory for Volterra and Abel integral equations of the second kind. *Math. Comp.* **1983**, *41*, 87–102. <https://doi.org/10.1090/S0025-5718-1983-0701626-6>.
22. Diethelm, K.; Ford, N.J.; Freed, A.D. Detailed error analysis for a fractional Adams method. *Numer. Algorithms* **2004**, *36*, 31–52. <https://doi.org/10.1023/B:NUMA.0000027736.85078.be>.
23. Diethelm, K. Smoothness properties of solutions of Caputo-type fractional differential equations. *Fract. Calc. Appl. Anal.* **2007**, *10*, 151–160.
24. Luchko, Y. Maximum principle for the generalized time-fractional diffusion equation. *J. Math. Anal. Appl.* **2009**, *351*, 218–223. <https://doi.org/10.1016/j.jmaa.2008.10.018>.
25. Luchko, Y.; Suzuki, A.; Yamamoto, M. On the maximum principle for the multi-term fractional transport equation. *J. Math. Anal. Appl.* **2022**, *505*, 125579. <https://doi.org/10.1016/j.jmaa.2021.125579>.
26. Al-Refai, M. On the fractional derivatives at extrema points. *Electron. J. Qual. Theory Differ. Equ.* **2012**, *55*, 1–5. <https://doi.org/10.14232/ejqtde.2012.1.55>.
27. Cong, N.D.; Tuan, H.T. Generation of nonlocal fractional dynamical systems by fractional differential equations. *J. Integral Equ. Appl.* **2017**, *29*, 585–608.
28. Tavazoei, M.S.; Haeri, M. A proof for non existence of periodic solutions in time invariant fractional order systems. *Automatica* **2009**, *45*, 1886–1890. <https://doi.org/10.1016/j.automatica.2009.04.001>.
29. Kaslik, E.; Sivasundaram, S. Non-existence of periodic solutions in fractional-order dynamical systems and a remarkable difference between integer and fractional-order derivatives of periodic functions. *Nonlinear Anal. Real. World Appl.* **2012**, *13*, 1489–1497. <https://doi.org/10.1016/j.nonrwa.2011.11.013>.
30. Baffet, D.; Hesthaven, J.S. A kernel compression scheme for fractional differential equations. *SIAM J. Numer. Anal.* **2017**, *55*, 496–520. <https://doi.org/10.1137/15M1043960>.
31. Jiang, S.; Zhang, J.; Zhang, Q.; Zhang, Z. Fast evaluation of the Caputo fractional derivative and its applications to fractional diffusion equations. *Commun. Comput. Phys.* **2017**, *21*, 650–678. <https://doi.org/10.4208/cicp.OA-2016-0136>.
32. Podlubny, I. *Fractional Differential Equations*; Academic Press: San Diego, CA, USA, 1999.
33. Hairer, E.; Wanner, G. *Solving Ordinary Differential Equations II. Stiff and Differential-Algebraic Problems*, 2nd ed; Springer Series in Computational Mathematics 14; Springer: Berlin/Heidelberg, Germany, 1996.
34. Flajolet, P.; Sedgewick, R. *Analytic Combinatorics*; Cambridge University Press: Cambridge, UK, 2009.
35. de Carvalho-Neto, P.M.; Fehlbeg Júnior, R. Conditions for the absence of blowing up solutions to fractional differential equations. *Acta Appl. Math.* **2018**, *154*, 15–29. <https://doi.org/10.1007/s10440-017-0130-5>.
36. Du, M.; Wang, Z.; Hu, H. Measuring memory with the order of fractional derivative. *Sci. Rep.* **2013**, *3*, 3431. <https://doi.org/10.1038/srep03431>.
37. Saeedian, M.; Khalighi, M.; Azimi-Tafreshi, N.; Jafari, G.R.; Ausloos, M. Memory effects on epidemic evolution: The susceptible-infected-recovered epidemic model. *Phys. Rev. E* **2017**, *95*, 022409. <https://doi.org/10.1103/PhysRevE.95.022409>.

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